

Primitive Carriers, Derivative Skins, and Classifier-Level Prediction Closure in the Standard Model

A Master Classifier for Same-Scope Primitive Exhaustion, New-Physics Sorting, and Carrier-Indexed Standing

Amos Jay Maley

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Abstract

This paper constructs the master classifier for a carrier-indexed account of Standard Model standing. The target is not empirical finality and not a prohibition on new physics. The target is classifier-level primitive discipline: no Standard Model-associated claim receives primitive standing merely by appearing as a field symbol, particle label, mass value, coupling, matrix entry, gauge choice, measurement, anomaly, effective-field-theory term, route, projection, or model name. We define the Standard Model claim arena D_{SM} , give a carrier-admission criterion, give non-circular skin-identification tests, introduce the component normal form

$$R_x = (D_{\text{SM}}, x, C_x, \text{Role}_x, \text{Rep}_x, \text{Meas}_x, \text{Transport}_x, \text{Route}_x, \text{Mix}_x, \text{Proj}_x, \\ \text{Extension}_x, \text{Update}_x, \text{Cert}_x, \rho_x, \text{Pred}_x),$$

and prove the SM-1 no-promotion results for objects that satisfy the skin-identification criteria without an independent carrier certificate. The paper also hardens target-update separation, burden status, carrier-certificate discipline, and non-ad-hoc classifiability. Future Standard Model-associated claims become classifier inputs rather than automatic endpoint escapees. Genuine new observations, anomalies, EFT displays, BSM proposals, and target-updating discoveries remain admissible for audit; they do not thereby acquire unlicensed same-scope primitive standing. The no same-scope primitive escape result is stated here as a sector-relative theorem schema. Its project-wide closure belongs to SM-6, after the gauge, matter, charge/coupling/RG, flavor/mixing/generation, Higgs, BSM/EFT, and projection-interface closures are received.

The AASC/kernel dependency used by this classifier is not an overlay placed on physics. It is the minimal non-degeneracy condition under which physical claims can preserve standing, reference, sameness, comparison, prediction, and falsification. Thus classifier-level discipline is not a discretionary taxonomy: it records the conditions under which a same-scope Standard Model claim remains the same claim under representation, measurement, projection, anomaly, or attempted extension.

Keywords: AASC; Standard Model; carrier standing; primitive closure; derivative skin; Higgs bridge carrier; BSM audit; EFT; classifier-level prediction register; falsification protocol; gauge structure; representation response; target update; no-promotion theorem.

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1 Kernel Status: Non-Degeneracy, Not Overlay

The AASC kernel is used here in its upstream role. It is not an interpretive overlay placed on physics, and it is not a discretionary taxonomy imposed after the physical content is fixed. It states the minimal non-degeneracy conditions under which a physical claim can retain standing, reference, sameness, comparison, prediction, and falsification at all. Without such conditions, the expressions “same claim”, “same target”, “same regime”, “same comparison”, “same prediction”, and “same failure” lose determinate force.

For this reason, SM-1 is not a vocabulary substitution for ordinary physics. It is a classifier-level primitive-status discipline theorem over kernel-preserved claimhood. The familiar objects in the classifier arena may be physically real, measurable, predictive, and indispensable in calculation. That does not by itself make them primitive carriers. Primitive standing requires a same-scope role that remains standing-bearing under the admissible transformations of the fixed target.

Definition 1.1 (Non-degenerate classifier claim). A Standard Model classifier claim is non-degenerate only when its target, reference-bearing status, admissible comparison class, and standing conditions are preserved. In notation,

$$\text{NonDegenerateClaim}_{\text{classifier}}(x) \Rightarrow \text{KernelPreserved}_{\text{classifier}}(x).$$

The implication is not a convention of this paper; it records the necessity condition for the claim to remain an identifiable object of assertion, prediction, comparison, or failure.

Theorem 1.2 (Sameness requires kernel preservation). *For SM-1 claims in D_{SM} , sameness is kernel-preserved sameness:*

$$\text{SameClaim}_{\text{classifier}}(x, y) \Rightarrow \text{KernelPreserved}_{\text{classifier}}(x, y).$$

Consequently,

$$\neg \text{KernelPreserved}_{\text{classifier}}(x, y) \Rightarrow \neg \text{SameClaim}_{\text{classifier}}(x, y).$$

A non-preserved target, comparison class, admissibility rule, or reference-bearing status yields a replacement, degeneration, target update, or failed comparison rather than the same claim under test.

Theorem 1.3 (Prediction and falsification require non-degenerate reference). *Every prediction or falsification claim in the classifier arena presupposes preserved target identity, comparison class, and reference standing:*

$$\begin{aligned} & \text{Prediction}_{\text{classifier}}(x) \vee \text{Falsification}_{\text{classifier}}(x) \\ \Rightarrow & \text{TargetPreserved}(x) \wedge \text{ComparisonClassPreserved}(x) \wedge \text{ReferenceStanding}(x). \end{aligned}$$

A prediction is a prediction of the same claim only under sameness preservation. A falsification is a falsification of the same claim only under target-preserving failure.

Proposition 1.4 (Kernel-denial burden). *A denial of the kernel is not a successful rejection of SM-1 unless it recovers the functions that the kernel supplies. Formally, any denial must satisfy the burden*

$$\begin{aligned} \text{KernelDenied}(x) \Rightarrow & \text{RecoverSameness}(x) \wedge \text{RecoverStanding}(x) \\ & \wedge \text{RecoverReference}(x) \wedge \text{RecoverFalsification}(x). \end{aligned}$$

Thus a classifier counterexample must preserve the same target, same comparison class, and same reference-bearing object while showing that carrier-admission, skin-identification, target-update, burden, prediction, and falsification statuses can be given without the kernel conditions used here.

Remark 1.5 (Proof class). The result is not a Lagrangian reconstruction, not a numerical parameter derivation, not an empirical-finality theorem, and not a no-new-physics theorem. It is a same-scope primitive-status exhaustion theorem for the stated arena. New measurements, anomalies, effective descriptions, and target-changing discoveries remain admissible; they do not become unlicensed same-scope primitive carriers merely by being real or useful.

2 Reader Orientation and Claim Boundary

2.1 Why this paper exists

The Higgs carrier arc exposed a method that is not restricted to the Higgs sector. A Standard Model component should not be treated as primitive merely because one can write it as a field, name it as a particle, measure a value, fit a coupling, choose a gauge, list a matrix entry, observe a detector signature, or display it through an EFT operator. Those displays can be valid; the question is whether they carry primitive standing. The answer is determined by the standing-bearing carrier role that licenses the display.

SM-1 generalizes the Higgs method into the master paper for the Standard Model carrier-exhaustion program. It does not attempt to finish the full sectoral closure. It constructs the status calculus and classifier that the later sector papers use. The central operational question is:

what licenses the familiar description as a standing-bearing SM claim?

The proposed answer is not a new phenomenological model and not a replacement Lagrangian. It is a classifier for the status of Standard Model-associated claims under a fixed comparison arena.

2.2 One-sentence thesis

The primitive Standard Model is its carrier network; the familiar Standard Model is the licensed skin atlas of that network.

Equivalently, the Standard Model is not primitively a list of fields and particles. It is a carrier-indexed standing architecture whose familiar fields, particles, parameters, measurements, and routes are licensed derivative descriptions.

2.3 Primitive closure versus empirical closure

The controlling distinction is between primitive closure and empirical closure. Primitive closure says that no same-scope Standard Model primitive remains outside the carrier network. Empirical closure would say that no new observations, anomalies, measurements, EFT descriptions, routes, or target-updating discoveries can occur. This project claims the former and rejects the latter.

The primitive-closure target is

$$\text{Survprim}_{D_{\text{SM}}}(\text{Attr}_{\text{SM}} \setminus C_{\text{SM}}) = 0.$$

The skin-openness condition is

$$\text{NewObservation}(y) \not\Rightarrow \text{NewPrimitive}(y).$$

New physics may be a skin novelty, route novelty, projection novelty, transport novelty, extension disposition, target update, burden state, or failed primitive. The point is not to reduce discovery. The point is to prevent novelty from being treated as primitive standing without a same-scope carrier certificate.

2.4 Nonclaims

Nonclaim 2.1 (What SM-1 does not claim). SM-1 does not claim that no future anomaly can occur, that no BSM model can be useful, that EFT language is dispensable, that detector signatures are irrelevant, that the complete Standard Model parameter table is derived, or that the final SM-6 endpoint theorem is already closed. It does not derive numerical gauge couplings, quark masses, CKM or PMNS values, Higgs potential coefficients, the Higgs vev, the Higgs scalar mass, scalar self-coupling, weak-scale numerical values, or absolute scale anchors.

Status 2.2 (What SM-1 claims). SM-1 claims a status calculus for Standard Model-associated claims; a no-promotion theorem for skins; target-update separation; future-claim classifiability; a prediction register and falsification protocol; a candidate carrier register; and a handoff architecture for SM-2 through SM-6. Final residual primitive deletion is reserved for SM-6 after the sector papers close the local carrier obligations.

2.5 Audit posture

The project is vulnerable if it is read as a conservative assertion that future physics is impossible. It is also vulnerable if its carrier language is read as a vocabulary substitution. SM-1 therefore fixes its audit posture at the start. A carrier is not a preferred description. A carrier must preserve standing under lawful redescription and must license familiar descriptions without being identical to any one display. A same-target objection has force only if it breaks a load-bearing theorem, invalidates the specialization to the Standard Model claim arena, or supplies a faithful same-scope primitive survivor outside the carrier network.

3 Running Example: The Higgs Carrier

The whole classifier can be read from the Higgs case before the full formal grammar is introduced. The Higgs arc separates the standing-bearing primitive from the familiar scalar-field skin. Its carrier is

$$C_H = H_{\text{br/fix}},$$

the Higgs bridge/fixation carrier received from the upstream Standard Model structural spine. Conventional Higgs scalar notation remains valid; it is simply not the primitive ground. The same holds for vev language, potential notation, pole-mass comparison, self-coupling values, lower-vacuum branch language, hierarchy comparisons, and extra-scalar model surfaces.

Submitted Higgs object	SM-1 status	Reason
Scalar field notation H	licensed representative	Field notation lawfully displays the Higgs sector but does not itself supply the bridge/fixation role.
vev	scale, transport, or measurement skin	A vev-like value belongs to an electroweak route and scale ledger, not to primitive Higgs standing.
Higgs potential	derivative route/status	Potential language is available inside the licensed scalar presentation; it is not the carrier.
pole mass	measurement witness	Pole-mass comparison witnesses a downstream excitation; it does not generate the primitive role.
self-coupling	parameter/transport target	Numerical self-coupling status is downstream unless separately certified.
lower vacuum branch	route, burden, target update, or failure	Branch availability is not primitive standing without same-scope carrier closure.
extra scalar	extension, burden, target update, or failure	A new scalar label must either descend, enrich, update the target, remain burdened, or fail primitive promotion.
hierarchy comparison	projection/scale-standing issue	Hierarchy language compares displays; it is not itself a primitive Higgs carrier.

This running example fixes the manuscript’s intended reading. SM-1 does not reject scalar-field phenomenology. It denies only primitive standing by scalar-field notation, vev language, potential notation, mass value, or extension label alone. The upstream structural spine treats $H_{\text{br/fix}}$ as structural while leaving Higgs potential, vev, scalar mass, self-coupling, weak-scale values, and numerical data downstream unless separately certified [2, 55, 56, 61].

4 Mathematical-Status Lock and Proof-Class Boundary

4.1 Proof class

This paper is an AASC-style exhaustion and status theorem over the Standard Model comparison arena. It is not a new phenomenological model, not a replacement for QFT calculations, not a computation of cross sections, and not a prediction of a new particle mass. Its proof class is classifier construction plus primitive-promotion deletion.

The core form is:

$$\text{ComponentClaim}_{\text{SM}}(x) \implies \rho_{\text{SM}}(x) \in \text{Status}_{\text{SM}}.$$

If a claim preserves the fixed target, it must receive a carrier-normal-form or registered downstream status. If it changes the target, it is not a same-scope survivor; it is a target update.

4.2 Dependency posture

This paper receives the upstream AASC and Standard Model structural machinery as load-bearing modules. The relevant burden is not whether the reader likes the terminology. The relevant burden is

whether the support chain fails at a theorem needed for the classifier. A standing objection must do at least one of the following:

1. break the AASC kernel or interface results used by the claim arena;
2. invalidate the physical-realization or Standard Model structural specialization;
3. show that a purported skin actually carries independent primitive standing under fixed D_{SM} ;
4. supply a faithful same-scope primitive survivor outside C_{SM} ;
5. show that the classifier requires ad hoc expansion to handle a real Standard Model-associated claim.

4.3 Relation to the standalone Standard Model proof spine

SM-1 receives the standalone Standard Model structural tuple as an upstream structural payload, not as a complete parameter table. The received tuple is

$$\Sigma_{\text{SM}} = (G_{\text{SM}}, \mathcal{R}_{\text{SM}}, H_{\text{br/fix}}, \mathcal{O}_Y, Q_{\text{SM}}, A_{\text{anom}}, \mathcal{V}_{\text{SM}}, M_{\text{slot}}, \Theta_{\text{mix}}, N_g) / \simeq_{\text{Adm}} .$$

Here G_{SM} is the recovered gauge-product survivor, $\mathcal{R}_{\text{SM}}$ is the chiral representation-response skeleton, $H_{\text{br/fix}}$ is the Higgs-like bridge/fixation carrier, $\mathcal{O}_Y$ is the Yukawa-realizability and coupling-slot ledger, Q_{SM} is the charge-lattice ledger, $A_{\text{anom}} = 0$ is the anomaly-neutral survivor ledger, $\mathcal{V}_{\text{SM}}$ is the vacuum-compatible residual normal form, M_{slot} and Θ_{mix} are mass-response and mixing/phase slots, and N_g is the family-cardinality envelope record [2].

The important boundary is that structural closure is not parameter closure. Gauge couplings, Yukawa eigenvalues, pole masses, CKM/PMNS coordinates, CP phases, Higgs potential, vev, scalar mass, self-coupling, weak-scale values, scale anchors, and dimensionless constants remain downstream unless a separate module supplies the relevant admissible witness, exactness certificate, reconstruction rule, quotient collapse, or zero-anchor derivation [2].

4.4 Why SM-1 comes before the sector papers

SM-1 supplies the master classifier and theorem family. SM-2 through SM-5 apply the classifier to sectors: gauge carriers, matter carriers, charge/coupling/RG transport, and flavor/mixing/generation. SM-6 receives those sector closures and assembles the final endpoint. Consequently, SM-1 must not pretend that every candidate carrier is already sector-proven. It records a candidate carrier register, marks proof-risk rows, and states the exact handoff obligations.

5 Theorem Status and Closure Level

5.1 Why status labels are load-bearing

SM-1 is the classifier paper, not the final exhaustion paper. Its results therefore have different closure levels. Some are definitions in the status grammar. Some are criteria for admitting a classifier verdict. Some are closed SM-1 theorems because they follow from the fixed classifier grammar and AASC standing discipline. Others are sector-relative schemas whose missing premise is local carrier closure in SM-2 through SM-5. The final residual primitive deletion theorem belongs to SM-6.

Status label	Meaning in SM-1	Later status if SM-6 closes
Definition	stipulated object in the classifier grammar, such as carrier, skin, target update, or burden	remains a definition
Criterion	admission or rejection test, such as carrier admission or skin identification	remains a criterion used inside theorems
Closed SM-1 theorem	follows from fixed classifier grammar, standing discipline, and no-promotion criteria	remains theorem
Sector-relative theorem schema	holds when the relevant sector closure is supplied by SM-2–SM-5	upgrades to theorem or corollary in SM-6
High-risk sector target	substantive local proof target, especially generation cardinality or same-scope BSM survivor tests	theorem if proved; otherwise burden/open row
SM-6 endpoint target	final project synthesis claim, not closed in SM-1	theorem only if SM-6 assembles the required closures

Principle 5.1 (No premature SM-6 closure).

$$\boxed{\text{SM-1} \neq \text{SM-6.}}$$

SM-1 proves classifier discipline, no primitive skin promotion under the skin-identification tests, target-update separation, burden non-survivor status, prediction-register typing, and the non-ad-hoc status grammar. SM-6 receives sector closures and proves the project-wide residual deletion theorem

$$\text{Survprim}_{D_{\text{SM}}}(\text{Attr}_{\text{SM}} \setminus C_{\text{SM}}) = 0.$$

5.2 Status-ranked theorem inventory

Result	SM-1 closure level	Upgrade condition
Carrier admission criterion	Criterion	remains criterion used by sector proofs
Skin identification lemma	Closed SM-1 lemma	remains lemma; sector papers apply it locally
No primitive skin theorem	Closed SM-1 theorem	remains theorem under fixed skin criteria
Classifier well-formedness	Closed SM-1 theorem	remains theorem
Classifier determinacy	Closed SM-1 theorem	remains theorem unless a real incompatible-status case is supplied
Non-ad-hoc classifiability	Closed SM-1 totality-or-failure theorem	future-claim classifiability if P0 survives grammar audit
Fixed-grammar non-ad-hoc classifiability	Corollary under $\text{GrammarAdequate}_{\text{SM}}$	remains a conditional theorem for the fixed grammar
No same-scope primitive escape	Sector-relative theorem schema	SM-6 theorem after sector closures

BSM audit closure	Conditional theorem schema	SM-6 theorem/corollary after BSM/EFT audit closures
Generation role-cardinality	High-risk sector target	theorem if SM-5 closes; otherwise explicit burden
SM carrier exhaustion closure	SM-6 endpoint target	final endpoint theorem only in SM-6

Remark 5.2 (Definitions do not upgrade). Definitions and criteria do not become theorems in SM-6. They remain the machinery by which SM-6 proves final theorems. Only schemas whose missing premises are local carrier closures upgrade to theorem level after those premises are supplied.

6 Technical Vocabulary and Standing Arena

6.1 Core terms

Definition 6.1 (Standing). Standing is admissible identity-bearing status for a construction, object, role, route, or witness inside a fixed scope. A claim with standing can be transported through admissible redescription without losing the object of comparison.

Definition 6.2 (Admissibility). Admissibility is the fail-closed permission to carry standing under declared scope, boundary, transport, and no-repair discipline. A move that destroys standing is not repaired by later usefulness.

Definition 6.3 (Reference). Reference is standing-preserving admissible redescription. A reference relation is not a mere notation change unless it preserves the fixed object under the active scope.

Definition 6.4 (Scope). Scope is the bounded domain in which admissibility, standing, redescription, target preservation, and carrier identity are evaluated.

Definition 6.5 (Primitive carrier). A primitive carrier is a standing-bearing role whose licensing relation remains invariant under admissible redescription. In the Standard Model arena, $\text{Carrier}_{\text{SM}}(x)$ means that x occupies a load-bearing role in the carrier network rather than merely appearing in a licensed display.

Definition 6.6 (Derivative skin). A derivative skin is a lawful field symbol, particle label, coordinate, basis, gauge choice, mass value, coupling value, matrix entry, measurement, route, projection, EFT display, or model-building surface whose validity depends on a carrier. The skin may be indispensable for calculation or communication; it is not primitive merely by being valid.

Definition 6.7 (Registered derivative statuses). The registered derivative statuses are licensed representative, measurement witness, transport ledger, route status, mixing status, projection display, and extension disposition. These statuses are not failures. They are the lawful downstream positions by which the familiar Standard Model remains available after primitive promotion is deleted.

6.2 Status atlas

The classifier returns a status

$$\rho_{\text{SM}}(y) \in \text{Status}_{\text{SM}},$$

$$\text{Status}_{\text{SM}} = \{\text{Carrier}, \text{LicensedRepresentative}, \text{MeasurementWitness}, \text{TransportLedger}, \text{RouteStatus}, \\ \text{MixingStatus}, \text{ProjectionDisplay}, \text{ExtensionDisposition}, \text{TargetUpdate}, \\ \text{CounterexampleBurden}, \text{FailedPrimitive}\}.$$

The status atlas is intentionally broad enough to receive known textbook components, new measurements, BSM proposals, EFT displays, and future anomalies. A future claim should not force ad hoc addition to the classifier merely because it is new.

Status	Meaning
Carrier	standing-bearing primitive role within D_{SM}
LicensedRepresentative	field, particle, notation, or conventional display licensed by a carrier
MeasurementWitness	empirical witness of carrier response or downstream route, not a primitive generator
TransportLedger	RG, scheme, basis, matching, calibration, normalization, or comparison ledger
RouteStatus	decay, scattering, confinement, oscillation, thermal, cosmological, detector, or phenomenological route
MixingStatus	transition-coordinate, basis, phase, matrix, or holonomy-like status
ProjectionDisplay	EFT, detector, metric, low-energy, Planck-ratio, event-topology, or model-projection display
ExtensionDisposition	carrier-preserving enrichment, role split, representative extension, or admissible BSM/EFT status
TargetUpdate	change of carrier inventory, invariant bundle, source state, admissibility boundary, or regime
CounterexampleBurden	unresolved same-scope non-reducing primitive claim with certificate still outstanding
FailedPrimitive	attempted primitive promotion from a skin, model name, route, number, measurement, or notation alone

6.3 Primitive carrier criteria

A proposed carrier must satisfy a strict identification discipline. It must specify a fixed target, a load-bearing role bundle, a licensing relation to derivative skins, redescription stability, no hidden derivative selector, the false primitive promotion being deleted, and the counterexample burden that would defeat it. This prevents the project from renaming a favored description as a carrier.

Criterion 6.8 (Carrier identification). A proposed C_x may be admitted as a candidate primitive carrier in D_{SM} only if it supplies:

1. a fixed same-scope target;
2. a load-bearing role bundle;
3. a licensing relation to ordinary field, particle, parameter, route, measurement, and projection descriptions;
4. stability under lawful changes of notation, gauge, basis, scheme, route, or projection;

5. absence of hidden selector import from a derivative skin;
6. the exact primitive promotion it deletes;
7. the same-scope survivor certificate that would falsify the classification.

7 The Standard Model Claim Arena D_{SM}

7.1 Definition of the arena

Definition 7.1 (Standard Model claim arena). The Standard Model claim arena is

$$D_{\text{SM}} = (\Sigma_{\text{SM}}, \text{Scope}_{\text{SM}}, \text{Inv}_{\text{SM}}, \text{RoleBundle}_{\text{SM}}, \text{Boundary}_{\text{SM}}, \text{Transport}_{\text{SM}}, \text{Admissibility}_{\text{SM}}).$$

It is the fixed same-scope arena in which Standard Model-associated claims are classified. It includes the recovered Standard Model structural tuple, the admissibility boundary, invariant bundle, role bundle, transport discipline, and target-preservation conditions.

The claim arena is not a complete numerical parameter table. It is the structural occupant and comparison discipline inherited from the standalone Standard Model proof spine. This prevents a parameter-value nonclosure from being misread as a structural failure, and it prevents a future measurement from being treated as primitive merely because it is empirically real.

7.2 Component claims

Definition 7.2 (Standard Model component claim). A Standard Model component claim is any submitted claim about a purported Standard Model object, role, parameter, field, route, measurement, anomaly, EFT term, or extension that presents itself as Standard Model-associated. Formally,

$$\text{ComponentClaim}_{\text{SM}}(x)$$

marks x as an input to the classifier.

The class includes ordinary claims such as “the Higgs is a scalar field,” “the photon is a gauge boson,” “a quark is a particle,” “the CKM matrix encodes flavor,” and “a coupling runs with scale.” It also includes future claims such as “a new resonance indicates a new scalar,” “a sterile neutrino has been found,” or “an EFT coefficient indicates new physics.”

7.3 Future and BSM claims

Definition 7.3 (Future Standard Model-associated claim). A future Standard Model-associated claim is any later experimental, theoretical, EFT, BSM, anomaly, or model-building claim asserted as connected to the Standard Model arena:

$$\text{FutureSMClaim}(y).$$

The classifier treats novelty as an input property, not as a standing certificate.

Definition 7.4 (BSM claim). A BSM claim is any proposed addition, extension, sector, threshold, field, symmetry, resonance, compositeness claim, unification structure, or hidden-sector proposal that relates itself to the Standard Model arena:

$$\text{BSMClaim}(b).$$

BSM status is audit-admissible, not primitive-protected.

7.4 Same-scope test

Definition 7.5 (Same-scope preservation). A claim y is same-scope under D_{SM} only if it preserves the carrier inventory, invariant bundle, source state, admissibility boundary, regime, role bundle, transport discipline, and no-hidden-load condition of D_{SM} :

$$\text{SameScope}_{D_{\text{SM}}}(y).$$

A claim fails the same-scope test when it changes the target while presenting itself as an internal addition. Many BSM proposals become apparent counterexamples only because they move the target while continuing to speak in Standard Model-associated language.

7.5 Target update

Definition 7.6 (Target update). A claim is a target update if it changes carrier inventory, invariant bundle, source state, admissibility boundary, regime, extension sector, gravitational/metric rule, source/measurement domain, or EFT threshold in a target-changing way:

$$\neg \text{SameScope}_{D_{\text{SM}}}(y) \Rightarrow \text{TargetUpdate}(y).$$

Target updates are not suppressed. They are structurally important. They are also not same-scope primitive survivors. This distinction is essential to preserving discovery while closing unlicensed primitive promotion.

8 Carrier Admission and Skin Identification

8.1 Carrier admission is not renaming

A carrier cannot be a preferred description. It must carry a role that remains stable when the lawful display changes. This section supplies the formal admission criterion that later sector papers must satisfy before any candidate row in the carrier register can be marked closed.

Criterion 8.1 (Carrier Necessary Conditions). For a Standard Model-associated object or role x , carrier standing requires the following necessary admission conditions:

$$\begin{aligned} \text{Carrier}_{\text{SM}}(x) \Rightarrow & \text{FixedTarget}(x) \wedge \text{RolePreservation}(x) \wedge \text{RedescriptionInvariant}(x) \\ & \wedge \text{LicensingRelation}(x) \wedge \text{NoHiddenSelector}(x) \wedge \text{BurdenDefined}(x). \end{aligned}$$

These conditions are necessary audit obligations, not by themselves a sector closure proof. A row that lacks them remains candidate, burdened, or failed.

Definition 8.2 (Closed carrier certificate). A closed carrier certificate is the necessary carrier-admission package plus a local closure proof:

$$\begin{aligned} \text{CarrierCertificate}_{\text{SM}}(x) := & \text{FixedTarget}(x) \wedge \text{RolePreservation}(x) \wedge \text{RedescriptionInvariant}(x) \\ & \wedge \text{LicensingRelation}(x) \wedge \text{NoHiddenSelector}(x) \wedge \text{BurdenDefined}(x) \wedge \text{LocalClosureProof}(x). \end{aligned}$$

SM-1 defines the certificate form. SM-2 through SM-5 close local carrier certificates. SM-6 assembles the certified carrier network.

Definition 8.3 (Non-circular content of local closure proof). The abbreviation $\text{LocalClosureProof}(x)$ does not mean “a proof that x is a carrier.” It abbreviates the local obligations that make the carrier certificate non-circular:

$$\begin{aligned} \text{LocalClosureProof}(x) &:= \text{RoleClosure}(x) \wedge \text{RedescriptionInvarianceProof}(x) \\ &\quad \wedge \text{LicensingProof}(x) \wedge \text{DescentExclusion}(x) \\ &\quad \wedge \text{NoHiddenSelectorProof}(x). \end{aligned}$$

Here $\text{RoleClosure}(x)$ proves closure of the role claimed by x , $\text{RedescriptionInvarianceProof}(x)$ proves invariance under the lawful changes relevant to that sector, $\text{LicensingProof}(x)$ proves the licensed relation to standard representatives and measurements, $\text{DescentExclusion}(x)$ proves that the claim is not merely a derivative skin or registered downstream status, and $\text{NoHiddenSelectorProof}(x)$ proves that no derivative display has been smuggled into primitive standing. Only after those local obligations close does the certificate suffice for carrier status.

Proposition 8.4 (Carrier Certificate Sufficiency). *A closed local certificate suffices for carrier status in the relevant sector:*

$$\text{CarrierCertificate}_{\text{SM}}(x) \Rightarrow \text{Carrier}_{\text{SM}}(x).$$

Carrier-admission field	Audit question
Fixed target	Which D_{SM} is held fixed while the claim is evaluated?
Role preservation	Which load-bearing role is preserved: gauge order, charge response, color transport, weak transition, matter response, flavor transition, Higgs fixation, mass response, or projection interface?
Redescription invariance	Does the role survive lawful changes of gauge, basis, scheme, normalization, notation, route, projection, or measurement context?
Licensing relation	Which fields, particles, values, matrices, routes, measurements, or projections are licensed by the carrier?
No hidden selector	Does the proposed carrier smuggle in a derivative skin as primitive selection principle?
Burden defined	What exact same-scope primitive survivor would defeat the carrier assignment?

The carrier certificate used in the worked examples is

$$\text{Cert}_C(x) = (D_{\text{SM}}, \text{Role}_x, \text{Inv}_x, \text{Lic}_x, \text{NoSelector}_x, \text{Falsifier}_x).$$

A sector paper closes a carrier only when this certificate is supplied and the relevant local closure proof is established for the role.

8.2 Skin identification is non-circular

The no-promotion theorem must not rely on the trivial inference that whatever is called a skin is nonprimitive. SM-1 therefore supplies operational skin-identification tests before applying the theorem.

Lemma 8.5 (Skin Identification Lemma). *A submitted object s is eligible for skin status only if at least one of the following is true: it is gauge-dependent, basis-dependent, scheme-dependent, normalization-dependent, measurement-route-dependent, projection-dependent, model-name-dependent, EFT-coordinate-dependent, downstream parameter-level, or lacks invariant role preservation under admissible redescription. In symbolic form,*

$$\begin{aligned} \text{Skins}_{\text{SM}}(s) &\Leftarrow \text{GaugeDep}(s) \vee \text{BasisDep}(s) \vee \text{SchemeDep}(s) \vee \text{NormDep}(s) \vee \text{MeasRouteDep}(s) \\ &\quad \vee \text{ProjDep}(s) \vee \text{ModelNameDep}(s) \vee \text{EFTCoordDep}(s) \vee \text{ParamStatus}(s) \vee \neg \text{RoleInvariant}(s). \end{aligned}$$

The arrow is intentionally one-way: satisfying one of these tests licenses skin classification; failing all of them does not by itself prove carrier status. Carrier status still requires the carrier-admission certificate.

Proof. Each listed condition identifies dependence on a display, coordinate, route, comparison, projection, model surface, or downstream value rather than invariant role preservation. Such dependence is compatible with lawful use but not with primitive standing unless an independent carrier certificate restores invariant role status. The lemma therefore classifies a disputed object by operational dependence rather than by vocabulary preference. $\square$

Theorem 8.6 (Non-Circular No-Promotion Form). *If s satisfies the skin-identification tests and lacks an independent carrier certificate, then s cannot be primitive in the Standard Model claim arena:*

$$\text{Skin}_{\text{SM}}(s) \wedge \neg \text{CarrierCertificate}(s) \Rightarrow \neg \text{Primitive}_{\text{SM}}(s).$$

Proof. By the Skin Identification Lemma, s depends on a display, route, coordinate, projection, model surface, measurement context, or downstream parameter status, or it fails invariant role preservation. Primitive standing requires role-stable preservation under admissible redescription. Without $\text{CarrierCertificate}(s)$, no independent carrier has been supplied to lift that dependence into primitive standing. Therefore s remains a lawful skin or derivative status and cannot be promoted to primitive status by its availability alone. $\square$

Remark 8.7 (Local application status). The Skin Identification Lemma supplies the SM-1 eligibility test for skin classification. Its application to particular Standard Model sectors is discharged locally in SM-2 through SM-5, except where an upstream arc, such as the Higgs carrier arc, already supplies the relevant certificate. Thus SM-1 proves the no-promotion form; sector papers prove the local skin-identification instances.

9 Primitive Carrier Versus Derivative Skin

9.1 Field symbols do not self-ground

A field symbol may be a valid representative without being a primitive carrier. The general deletion is

$$\text{FieldSymbol}(x) \not\Rightarrow \text{Carrier}(x).$$

The Standard Model frequently uses field notation because field notation is the appropriate calculational and representational skin. But field notation cannot bypass carrier identification. The Higgs arc supplies the prototype: the conventional scalar notation is a licensed representative of the bridge/fixation carrier, not the primitive Higgs ground itself [55, 56, 61].

9.2 Particle labels do not self-ground

Similarly,

$$\text{ParticleLabel}(x) \not\Rightarrow \text{Primitive}(x).$$

The labels “quark,” “gluon,” “photon,” “neutrino,” and “Higgs boson” can be indispensable; they do not settle primitive status. A quark label does not imply isolated-particle standing. A perturbative gluon label does not imply free asymptotic primitive standing. A sterile-neutrino label does not decide whether the claim is a route, extension, burden, or target update.

9.3 Numerical values do not self-ground

Masses, charges, couplings, running values, mixing angles, phase conventions, pole masses, vevs, and self-couplings are not primitive merely because they are numerical:

$$\text{Number}(n) \not\Rightarrow \text{Carrier}(n).$$

A number can be a measurement witness, calibration, transport coordinate, boundary display, ratio-level output, or downstream parameter target. Its status depends on the carrier and certificate that license it.

9.4 Measurements do not generate primitive standing

Measurement witnesses are crucial, but they do not generate primitive carrier status. A measurement can witness response, constrain a route, raise burden severity, or motivate target-update tests. It cannot create the primitive role by measurement alone:

$$\text{MeasurementWitness}(m, C_x) \not\Rightarrow \text{Primitive}(m).$$

This is not skepticism about measurement. It is a typing rule. Empirical standing and primitive standing are different statuses.

9.5 Routes and projections do not self-ground

Decay routes, scattering amplitudes, jets, oscillation baselines, cosmological bounds, detector events, EFT coefficients, RG plots, and low-energy displays are route or projection statuses. They can pressure the carrier network, but they do not become primitive merely by being useful or surprising. The classifier asks whether the route descends to a carrier, signals a target update, remains a burden, or fails primitive promotion.

Theorem 9.1 (No Primitive Skin Theorem). *Let s be a Standard Model-associated object that satisfies the skin-identification tests: a field symbol, particle label, mass value, charge number, coupling value, matrix entry, gauge choice, basis convention, measurement, route, projection, EFT display, model name, or downstream parameter display. Then*

$$\text{Skin}_{\text{SM}}(s) \wedge \neg \text{CarrierCertificate}(s) \Rightarrow \neg \text{Primitive}_{\text{SM}}(s).$$

If s is later shown to occupy a standing-bearing carrier role, it is admitted through the carrier test rather than through skin promotion.

Proof. The theorem uses the Skin Identification Lemma, not a verbal stipulation. A skin is identified by dependence on gauge, basis, scheme, normalization, measurement route, projection, model surface, EFT coordinate, downstream parameter status, or failure of invariant role preservation. Primitive standing requires invariant role preservation under fixed D_{SM} . Without a carrier certificate, the submitted object cannot supply that invariant role. Therefore the object may be lawful and useful, but it remains nonprimitive. If an independent carrier certificate is later supplied, the verdict changes by carrier admission, not by self-promotion of the skin. $\square$

10 Component Normal Form R_x

10.1 Definition

The component normal form is the audit object assigned to a submitted claim:

$$R_x = (D_{\text{SM}}, x, C_x, \text{Role}_x, \text{Rep}_x, \text{Meas}_x, \text{Transport}_x, \text{Route}_x, \text{Mix}_x, \text{Proj}_x, \\ \text{Extension}_x, \text{Update}_x, \text{Cert}_x, \rho_x, \text{Pred}_x).$$

The normal form is not a slogan. It is a record of the fixed target, carrier candidate, role, licensed representatives, measurement witnesses, transport ledgers, route statuses, mixing statuses, projection displays, extension dispositions, target-update flags, certificates, final classifier status, and prediction-register assignment.

10.2 Entry definitions

Entry	Meaning
D_{SM}	fixed Standard Model claim arena
x	submitted object or claim
C_x	candidate primitive carrier, if any
Role_x	load-bearing standing role
Rep_x	field, particle, notation, representation, or conventional display
Meas_x	measurement witness status
Transport_x	RG, scheme, basis, matching, calibration, normalization, or comparison ledger
Route_x	decay, scattering, confinement, oscillation, detector, thermal, or cosmological route
Mix_x	basis, phase, matrix, transition-coordinate, or holonomy-like status
Proj_x	EFT, metric, detector, low-energy, or phenomenological projection
Extension_x	BSM/EFT enrichment disposition
Update_x	target-update status
Cert_x	certificate or proof obligation
ρ_x	final classifier status
Pred_x	prediction-register assignment

10.3 Adequacy conditions

Criterion 10.1 (Normal-form adequacy). A component normal form R_x is adequate only if:

1. all entries preserve the fixed target or explicitly mark target update;
2. every derivative skin has a licensing carrier or receives burden/failure status;
3. no route, measurement, matrix, value, or notation is promoted into primitive standing without a carrier certificate;
4. counterexample burden is not counted as survivor status;
5. prediction status includes expected disposition, falsifier, evidence route, and current proof status.

10.4 Certificate field

The certificate field has the internal form

$$\text{Cert}_x = (\text{Input}_x, \text{RolePres}_x, \text{Descent}_x, \text{NonPromotion}_x, \text{Falsifier}_x, \text{Status}_x).$$

It records the submitted input, role preservation condition, descent result, non-promotion deletion, falsifier, and current status. A certificate may close the claim, mark it as sector-proven, leave it as burden, or fail primitive promotion.

11 The Master Classifier

11.1 Classifier overview

The master classifier is

$$\mathcal{C}_{\text{SM}} : \text{Claim}_{\text{SM}} \rightarrow \text{Status}_{\text{SM}}.$$

It receives known textbook components and future claims. Its output is $\rho_{\text{SM}}(x)$, together with the normal form R_x . The classifier is designed to prevent two mistakes: first, treating valid descriptions as primitive without carrier proof; second, treating target-changing discoveries as same-scope counterexamples.

11.2 Stage 1: target test

The first question is whether the claim preserves D_{SM} , carrier inventory, invariant bundle, source state, admissibility boundary, regime, role bundle, and transport discipline. If not,

$$\rho_{\text{SM}}(x) = \text{TargetUpdate}.$$

This stage is deliberately first. A claim cannot falsify same-scope primitive closure by changing the target and then declaring itself an internal Standard Model primitive.

11.3 Stage 2: carrier test

If the claim preserves the target, the classifier asks whether it occupies a standing-bearing carrier role with a preserved licensing relation to standard descriptions. If yes,

$$\rho_{\text{SM}}(x) = \text{Carrier}.$$

The carrier test is strict. A field symbol, matrix, mass value, symmetry name, or model label cannot pass merely by being familiar or useful.

11.4 Stage 3: descent test

If the claim is not a carrier, the classifier asks whether it descends to an existing carrier as representative, measurement, transport, route, mixing, projection, extension disposition, or skin novelty. If yes, the appropriate derivative status is assigned. Descent is a success condition, not a demotion in ordinary physics. It says that the claim is licensed by the carrier network.

11.5 Stage 4: prediction test

If the claim is future, anomalous, proposed, BSM, or model-building in character, the classifier checks the Prediction Register. If governed by a register item, $Pred_x = P_i$. This stage makes prediction a formal project artifact rather than an informal consequence.

11.6 Stage 5: burden test

If a same-scope claim asserts non-reducing primitive standing but does not close the certificate, it receives counterexample-burden status:

$$\rho_{SM}(x) = \text{CounterexampleBurden}.$$

Burden is not survivor status. It is a review state.

11.7 Stage 6: failure test

If the claim asserts primitive standing while supplying only a field symbol, particle label, measurement, route, matrix entry, parameter value, model name, symmetry name, or projection display, it fails primitive promotion:

$$\rho_{SM}(x) = \text{FailedPrimitive}.$$

Proposition 11.1 (Fixed-Grammar Status Coverage). *Under the declared status grammar and adequate fixed inputs, each submitted Standard Model-associated claim receives a classifier status:*

$$\text{ComponentClaim}_{SM}(x) \wedge \text{GrammarAdequate}_{SM}(x) \Rightarrow \exists \rho_{SM}(x) \in \text{Status}_{SM}.$$

This proposition is a fixed-grammar coverage result, not the final future-claim theorem. A future claim that defeats the grammar registers as failure of P0 rather than as an ad hoc new status.

Proof. The target test partitions claims into target-preserving and target-updating. For target-updating claims, **TargetUpdate** is assigned under the target-update criterion. For target-preserving claims, the carrier test either closes a carrier certificate or passes the claim to the descent test. Descent checks the finite registered derivative atlas. If descent does not close but a same-scope primitive survivor is asserted, burden status is assigned until a certificate closes. If the claim supplies only a skin while asserting primitive standing, failed-primitive status is assigned. These stages exhaust the fixed grammar when $\text{GrammarAdequate}_{SM}(x)$ holds. $\square$

12 Non-Ad-Hoc Classifiability and Classifier Closure

12.1 Why classifiability is not a catch-all maneuver

Classifiability would be trivial if **TargetUpdate**, **CounterexampleBurden**, and **FailedPrimitive** were arbitrary residual bins. They are not. They are rule-governed terminal statuses with fixed entry conditions and fixed escalation paths. A future claim may require a new row in the prediction register, but it may not receive a new primitive status category unless the classifier grammar itself fails. Such a failure is not hidden; it is the falsifier for P0.

Definition 12.1 (Ordered status grammar). The classifier uses a finite ordered status grammar:

$$T_{\text{target}} \prec T_{\text{carrier}} \prec T_{\text{descent}} \prec T_{\text{prediction}} \prec T_{\text{burden}} \prec T_{\text{failure}}.$$

Earlier stages have priority when their input conditions close. Later stages are reached only after the earlier stage has either failed or assigned a compatible ancillary tag.

Theorem 12.2 (Classifier Well-Formedness). *Every Standard Model-associated claim is evaluated by a finite ordered status grammar:*

$$\text{Claim}_{\text{SM}}(x) \Rightarrow \text{EvaluatedByFiniteGrammar}(x, \text{Status}_{\text{SM}}).$$

Proof. The status set is finite and explicitly listed. The stage order is finite. Each stage has an entry condition: target preservation, carrier certificate, derivative descent, prediction-register match, same-scope primitive assertion with incomplete certificate, or primitive assertion from skin alone. No stage introduces an unlisted primitive status. Therefore the classifier is well-formed. $\square$

Theorem 12.3 (Classifier Determinacy). *For fixed inputs, the classifier does not assign incompatible terminal statuses. If two candidate statuses appear, either they are compatible ancillary tags or the ordered grammar selects the unique terminal status:*

$$\text{FixedInputs}(x) \wedge \rho_i(x) \wedge \rho_j(x) \Rightarrow \text{Compatible}(\rho_i, \rho_j) \vee \text{PrioritySelects}(\rho_i, \rho_j).$$

Proof. Target update has priority over same-scope statuses because a target-changing claim is not evaluated as a same-scope survivor. Carrier status has priority over derivative descent because a closed carrier certificate supplies primitive role standing rather than downstream display. Descent statuses may coexist as ancillary tags, for example measurement plus route plus projection, but the terminal verdict records the highest relevant status. Burden applies only when carrier closure is asserted but not certified and descent/target update have not settled the claim. Failed primitive applies only when the primitive assertion rests on a skin without carrier certificate. Thus incompatible statuses are blocked by entry conditions and priority order. $\square$

Lemma 12.4 (Non-Ad-Hoc Extension). *A future claim may require a new prediction-register row, worked certificate, or sector handoff, but it may not require a new primitive status category without falsifying universal future-claim sorting:*

$$\text{FutureSMClaim}(y) \wedge \neg \exists \rho_{\text{SM}}(y) \Rightarrow \text{Failure}(P0).$$

Proof. The classifier is committed in advance to the listed statuses. If a real Standard Model-associated claim cannot be assigned any of those statuses after fixed inputs, the correct response is not to create an ad hoc status, but to mark P0 as failed or the classifier grammar as incomplete. This turns classifiability into a falsifiable commitment. $\square$

Theorem 12.5 (Non-Ad-Hoc Classifiability). *Given fixed inputs and the ordered status grammar, every submitted Standard Model-associated claim either receives a terminal status or falsifies P0 by requiring grammar expansion:*

$$\text{Claim}_{\text{SM}}(x) \Rightarrow (\exists \rho_{\text{SM}}(x) \in \text{Status}_{\text{SM}}) \vee \text{Failure}(P0).$$

SM-1 does not claim that no conceivable future case could pressure the grammar. It claims that such pressure is visible as failure of P0, not hidden by ad hoc status invention.

Proof. The target test first separates target updates from same-scope claims. Same-scope claims then either close a carrier certificate, descend to a registered derivative status, match a prediction-register row, assert a same-scope primitive with certificate still open, or rely on a skin alone. These are exhaustive for the fixed grammar. If a real claim satisfies none of the entry conditions and no status can be assigned without adding a new terminal category, P0 fails. Therefore classifiability is non-ad hoc: it is either achieved by the fixed grammar or explicitly falsified. $\square$

Corollary 12.6 (Fixed-Grammar Classifier Totality). *If the declared grammar is adequate for the submitted claim, the left disjunct closes:*

$$\text{Claim}_{\text{SM}}(x) \wedge \text{GrammarAdequate}_{\text{SM}}(x) \Rightarrow \exists \rho_{\text{SM}}(x) \in \text{Status}_{\text{SM}}.$$

13 Main Theorem Family

Theorem 13.1 (SM Carrier Normal Form Theorem). *Every same-scope standing-bearing Standard Model component claim factors through a carrier normal form or receives a registered downstream status:*

$$\text{ComponentClaim}_{\text{SM}}(x) \wedge \text{SameScope}_{D_{\text{SM}}}(x) \Rightarrow \exists R_x [\text{NormalForm}_{\text{SM}}(R_x)].$$

Proof. By fixed-grammar status coverage, x receives a status when the declared grammar is adequate; otherwise P0 fails rather than expanding the status set ad hoc. If x is a carrier, C_x and Role_x are nonempty in the normal form. If x is derivative, it receives a registered downstream entry in $\text{Rep}_x, \text{Meas}_x, \text{Transport}_x, \text{Route}_x, \text{Mix}_x, \text{Proj}_x$, or Extension_x . If x asserts primitive status but lacks certificate closure, it is entered as burden. If it relies only on skin promotion, it is entered as failed primitive. In every case, the entries required for R_x are determined by the classifier. $\square$

Theorem 13.2 (Measurement Witness Non-Promotion). *A measurement can witness carrier response but cannot create primitive carrier standing:*

$$\text{MeasurementWitness}(m, C_x) \Rightarrow \text{Witness}(m, C_x) \wedge \neg \text{Primitive}(m).$$

Proof. Measurement standing is relational: it records response under an admissible route and comparison discipline. If measurement generated primitive standing, empirical detection would bypass the carrier test and would allow target-preserving primitive addition from route availability alone. That contradicts the distinction between witness and carrier. Measurement may raise burden severity or trigger target-update tests; it does not itself occupy the carrier role. $\square$

Theorem 13.3 (Transport Non-Generation). *RG flow, matching, scheme conversion, basis change, calibration, and normalization preserve or route standing; they do not generate primitive standing:*

$$\text{TransportLedger}(t, C_x) \Rightarrow \text{PreservesOrRoutes}(t, C_x) \wedge \neg \text{GeneratesPrimitive}(t).$$

Proof. A transport ledger compares or carries standing across lawful descriptions. If the transport object itself generated primitive standing, scheme, basis, or matching choices could become primitive by chart selection. A ledger can identify invariant residue, reveal obstruction, or show mismatch; it cannot promote the coordinate of comparison into the primitive object. $\square$

Theorem 13.4 (Target Update Separation). *If a claim changes carrier inventory, source state, invariant bundle, admissibility boundary, role bundle, or regime, then it is a target update rather than a same-scope survivor:*

$$Claim(y) \wedge \neg SameScope_{D_{SM}}(y) \Rightarrow TargetUpdate(y),$$

and

$$TargetUpdate(y) \Rightarrow \neg SameScopeSurvivor(y).$$

Proof. Same-scope survivor status is defined relative to fixed D_{SM} . A claim that changes the arena cannot preserve the target required for same-scope comparison. It may be empirically decisive and may open a new theory domain, but it does so by updating the target rather than by surviving inside the old one. $\square$

Theorem 13.5 (Future-Claim Classifiability). *Every future Standard Model-associated claim is a classifier input rather than an endpoint escape:*

$$FutureSMClaim(y) \Rightarrow \exists \rho_{SM}(y).$$

Proof. Future status changes chronology, not type. A future claim is submitted as measurement, route, projection, EFT term, model, field, symmetry, particle label, threshold, anomaly, or carrier assertion. These are all handled by the target, carrier, descent, prediction, burden, and failure tests. If the claim changes the target, target-update status applies. If it preserves the target, the remaining classifier stages apply. $\square$

Theorem 13.6 (No Same-Scope Primitive Escape Schema). *For any future claim whose relevant sector carrier closure is locally established,*

$$FutureSMClaim(y) \wedge SameScope_{D_{SM}}(y) \Rightarrow \neg Survprim_{D_{SM}}(y \setminus C_{SM}).$$

Proof schema. Assume y is same-scope. By future-claim classifiability, y receives classifier status. If y is a carrier, it belongs to the locally established carrier closure and is not outside C_{SM} . If it descends, it is derivative. If it is burdened, it is not a survivor. If it fails primitive promotion, it is not primitive. Since target update is excluded by same-scope assumption, no remaining same-scope primitive survivor exists outside the closed sector carrier network. Full project-wide closure is the SM-6 endpoint after sector closures are received. $\square$

Theorem 13.7 (BSM Audit Closure). *For any BSM claim,*

$$\begin{aligned} & BSMClaim(b) \wedge SameScope_{D_{SM}}(b) \\ \Rightarrow & Descends(b) \vee RegisteredDerivative(b) \vee CounterexampleBurden(b) \vee FailedPrimitive(b). \end{aligned}$$

If the BSM claim is not same-scope, then

$$BSMClaim(b) \wedge \neg SameScope_{D_{SM}}(b) \Rightarrow TargetUpdate(b).$$

Proof. A same-scope BSM claim is still a claim inside D_{SM} . It is therefore subject to the carrier and descent tests. If it supplies no closed primitive carrier certificate outside the carrier network, it descends, remains burdened, or fails. If it changes the carrier inventory, regime, invariant bundle, or admissibility boundary, it is not same-scope and is classified as target update. $\square$

Theorem 13.8 (Counterexample Burden Is Not Survivor).

$$CounterexampleBurden(y) \Rightarrow \neg PrimitiveSurvivor(y).$$

Proof. Burden status records an unresolved certificate obligation. Survivor status requires closure of the same-scope primitive-standing certificate after target update and derivative statuses have been excluded. Since burden is precisely the absence of that closure, it is not survivor status. $\square$

14 Target Update, Burden Status, and Falsifiability

14.1 Target update is not counterexample avoidance

Target update is a positive structural status, not a refusal to count evidence. A claim does not become a target update merely because it is surprising, new, BSM-labeled, statistically significant, or phenomenologically useful.

The following do not automatically count as target updates: new particle names, new field notation, new scalar labels, new resonances, new anomalies, new EFT coefficients, new fitted parameters, new symmetry proposals, new unification plots, new decay routes, or new detector topologies. These first enter the classifier as claims requiring target, carrier, descent, prediction, burden, or failure tests.

A claim becomes a target update only if it changes at least one load-bearing feature of the comparison arena: carrier inventory, invariant bundle, admissibility boundary, source state, role bundle, comparison arena, regime conditions, load-bearing transport rules, or primitive domain.

Criterion 14.1 (Target-Update Discipline Criterion).

$$\begin{aligned} \text{TargetUpdate}(y) \Rightarrow & \text{ChangedCarrierInventory}(y) \vee \text{ChangedInvariantBundle}(y) \vee \text{ChangedBoundary}(y) \\ & \vee \text{ChangedSourceState}(y) \vee \text{ChangedRoleBundle}(y) \vee \text{ChangedArena}(y). \end{aligned}$$

Thus target-update status requires an independently stated target-changing condition.

This is a classification criterion rather than an independent sector theorem. Its force is restrictive: it prevents target-update status from being used unless an independently stated target-changing condition is present. By definition, same-scope status is evaluated only when the target, carrier inventory, invariant bundle, boundary, source state, role bundle, and comparison arena are held fixed. A claim that changes none of these is still inside D_{SM} and cannot be classified as target update merely because it is disruptive.

14.2 Burden state is review state, not survivor state

Definition 14.2 (Counterexample burden). $\text{CounterexampleBurden}(y)$ means that y asserts same-scope primitive standing, has not supplied a closed carrier certificate, has not fully defeated derivative descent, and has not fully excluded target-update status. Burden is a live audit state. It is neither acceptance nor dismissal.

Theorem 14.3 (Burden Escalation Criterion). *A burden can escalate to primitive-survivor status only if*

$$\begin{aligned} & \text{SameScope}_{D_{\text{SM}}}(y) \wedge \text{OutsideCarrierNetwork}(y) \wedge \text{NonDerivative}(y) \\ & \wedge \text{CarrierCertificate}(y) \wedge \text{ClosedCertificate}(y). \end{aligned}$$

Without those conditions,

$$\text{CounterexampleBurden}(y) \Rightarrow \neg \text{PrimitiveSurvivor}(y).$$

Proof. The conditions state exactly what burden lacks: fixed same-scope preservation, proof that the claim lies outside the carrier network, exclusion of derivative statuses, and a closed primitive carrier certificate. Until those are supplied, the claim remains a burden rather than a survivor. $\square$

14.3 A counterexample that cannot hide as target update

Let y^* be a future claim that preserves D_{SM} , preserves the carrier inventory, preserves the invariant bundle, preserves the role bundle, is not a measurement, route, projection, EFT coordinate, model name, matrix coordinate, or transport ledger, and supplies a closed primitive carrier certificate outside C_{SM} . Then y^* is not a target update and not a failed primitive. It is a genuine same-scope primitive survivor. If certified, it defeats the same-scope primitive-exhaustion endpoint.

This example is deliberately hostile. Its function is to show that the classifier has a real defeat condition and does not reclassify every pressure case away. If y^* closes, then

$$y^* \Rightarrow \text{Failure}(P1),$$

and, if SM-6 had asserted final residual primitive deletion, also

$$y^* \Rightarrow \text{Failure}(\text{SM6 Endpoint}).$$

The classifier is not allowed to reclassify y^* as target update unless one of the target-update criteria is satisfied, and it is not allowed to leave y^* as burden once the closed carrier certificate is supplied.

14.4 Target-update success case

Suppose a future discovery z supplies an independently standing carrier, cannot descend to any member of C_{SM} , changes the invariant bundle, changes the carrier inventory, carries a closed certificate, and preserves admissibility under a new target D' . Then

$$\rho_{\text{SM}}(z) = \text{TargetUpdate},$$

not $\text{FailedPrimitive}(z)$. A target update is not a failed claim. It is a structurally successful discovery whose standing is not same-scope Standard Model standing.

15 New Physics Diagnostic Architecture

15.1 Why new physics is not one status

In ordinary usage, “new physics” can mean a measurement anomaly, a fitted deviation, an EFT coefficient, a new resonance, a hidden sector, a new symmetry, a threshold, compositeness, a gravity-coupled regime, or a primitive carrier change. These are not one status. A classifier that treats all novelty as primitive will overpromote; a classifier that treats all novelty as failure will suppress discovery. The master classifier does neither.

15.2 Diagnostic table

New-physics appearance	Classifier question	Possible status
Anomaly	Is it measurement deviation, route stress, projection mismatch, transport correction, burden state, or target update?	MeasurementWitness / RouteStatus / ProjectionDisplay / TransportLedger / Burden / TargetUpdate
Resonance	Is it detector skin, route enrichment, carrier-preserving extension, or new carrier requiring target update?	ProjectionDisplay / RouteStatus / ExtensionDisposition / Burden / TargetUpdate
EFT coefficient	Is it projection/transport ledger or evidence of changed target conditions?	ProjectionDisplay / TransportLedger / Burden / TargetUpdate
New symmetry	Does it supply a carrier role or only a model-building skin?	Carrier / LicensedRepresentative / FailedPrimitive / TargetUpdate
New scalar	Is it Higgs carrier split, extension disposition, or target update?	ExtensionDisposition / Burden / TargetUpdate / FailedPrimitive
Fourth generation	Does it satisfy role-cardinality closure under fixed carrier inventory?	Burden / FailedPrimitive / TargetUpdate
Sterile neutrino	Is it weak-flavor route, independent carrier, or target change?	RouteStatus / ExtensionDisposition / Burden / TargetUpdate
Dark photon	Is it projection/interface enrichment or new electromagnetic carrier?	ProjectionDisplay / ExtensionDisposition / Burden / TargetUpdate

15.3 Discovery-positive posture

The classifier is discovery-positive because it specifies what kind of newness has been found. A new observation becomes more informative when it can be distinguished as a measurement witness, route stress, projection novelty, transport update, extension disposition, burden state, or target update. The project denies only unlicensed same-scope primitive promotion.

Theorem 15.1 (New-Physics Diagnostic Theorem). *For every Standard Model-associated new-physics claim y ,*

$$\begin{aligned}
 & \text{NewPhysics}(y) \Rightarrow \rho_{\text{SM}}(y) \in \text{Status}_{\text{SM}}, \\
 & \text{Status}_{\text{SM}} = \{\text{Carrier, LicensedRepresentative, MeasurementWitness, TransportLedger, RouteStatus,} \\
 & \quad \text{MixingStatus, ProjectionDisplay, ExtensionDisposition, TargetUpdate,} \\
 & \quad \text{CounterexampleBurden, FailedPrimitive}\}.
 \end{aligned}$$

A real claim that occupies no registered status after fixed inputs would falsify P0 and the non-ad-hoc classifiability commitment.

16 Prediction Register and Falsification Protocol

16.1 Target-preserving prediction and falsification

The prediction register in SM-1 has reality contact only because the target and comparison class are preserved. It is not an internal bookkeeping list. A predicted measurement, anomaly, route, or extension pressure is assessed as a prediction of the same Standard Model classifier claim only when the reference-bearing target in D_{SM} remains fixed.

Proposition 16.1 (Reality-contact discipline for classifier predictions). *For every classifier prediction or falsification entry,*

$$\begin{aligned} & \text{Prediction}_{\text{classifier}}(x) \vee \text{Falsification}_{\text{classifier}}(x) \\ \Rightarrow & \text{KernelPreserved}_{\text{classifier}}(x) \wedge \text{TargetPreserved}(x) \wedge \text{ComparisonClassPreserved}(x). \end{aligned}$$

If the target changes, the result is a target update. If the target is preserved but the alleged primitive status does not close, the result is a registered status, burden, or failed primitive promotion rather than a same-scope primitive survivor.

16.2 Prediction-layer thesis

The prediction layer is not a list of numerical mass predictions. It is a family of primitive-status predictions over the future status of Standard Model-associated claims:

$$\begin{aligned} & \text{FuturePhysics}(y) \Rightarrow \text{Skin/Route/Measurement/Projection/Transport/} \\ & \text{Extension/TargetUpdate/Burden/Failure,} \end{aligned}$$

not

$$\text{New}(y) \Rightarrow \text{Primitive}(y).$$

This is testable because it states what would falsify it: a same-scope primitive survivor outside the carrier network with derivative statuses and target update excluded.

16.3 Two prediction classes

The register contains two different kinds of forward commitment. P0–P3 are classifier-risk commitments: their failure pressures the classifier architecture itself. P4–P10 are sector primitive-status predictions: their failure pressures a sector closure, local carrier admission, or local skin-identification result.

$$\text{ClassifierRiskCommitment}(P_i) \quad (i = 0, 1, 2, 3),$$

$$\text{SectorClassificationPrediction}(P_i) \quad (i = 4, 5, 6, 7, 8, 9, 10).$$

This distinction prevents equivocation between a grammar-level risk and a local sector prediction.

16.4 General prediction theorem family

Prediction theorem	Statement	Failure condition
Future-Claim Classifiability	Every future SM-associated claim receives classifier status after fixed inputs.	A real future claim cannot be classified without ad hoc expansion.
No Same-Scope Primitive Escape	Same-scope future claims do not survive as primitive outside C_{SM} , relative to closed sector carriers.	A same-scope future primitive closes outside C_{SM} .
New-Physics Diagnostic Dichotomy	New physics sorts into registered statuses.	A real same-scope claim occupies no registered status.
BSM Audit Closure	Same-scope BSM claims descend, remain burdens, or fail primitive promotion.	Same-scope BSM primitive standing closes outside C_{SM} .
Target-Update Separation	Target-changing claims are updates, not same-scope survivors.	A target-changing claim is counted as a same-scope survivor.

16.5 Project-wide Prediction Register

Table 2: SM-1 Prediction Register

ID	Type	Prediction	Expected classification	Falsifier
P0	classifier-risk commitment	Universal future-claim sorting	Every future SM-associated claim receives one of the fixed statuses, or P0 fails.	A real claim cannot be classified without grammar expansion.
P1	classifier-risk commitment	No same-scope BSM primitive escape	Same-scope BSM additions descend, reclassify, remain burdened, or fail primitive promotion.	A BSM claim preserves D_{SM} and supplies independent primitive standing outside C_{SM} .
P2	classifier-risk commitment	Skin-level new physics remains possible	Observations, anomalies, EFT displays, and routes may be real without becoming primitives.	A new observation requires primitive standing while preserving D_{SM} and defeating descent.
P3	classifier-risk commitment	Target-updating new physics is separated	Target-changing discovery is classified as target update, not same-scope survivor.	A target-changing discovery is validly counted as same-scope primitive.

ID	Type	Prediction	Expected classification	Falsifier
P4	sector prediction	Higgs-sector deviations	Deviations classify as measurement, route, projection, extension, target update, burden, or failed primitive.	A same-scope Higgs primitive outside $C_H = H_{\text{br/fix}}$ is forced.
P5	sector prediction	Quark confinement standing	Quarks do not obtain primitive isolated-particle standing under fixed SM scope.	A free quark with same-scope primitive SM standing is established.
P6	sector prediction	Gluon non-asymptotic standing	Gluons classify as non-asymptotic color-transport standing; jets and color-flow diagrams are derivative statuses.	A free gluon with same-scope primitive SM standing is established.
P7	sector prediction	Coupling/RG transport	Coupling values, running crossings, matching conditions, and scheme coordinates are transport/calibration ledgers.	An RG crossing or bare coupling value generates primitive standing without carrier normal form.
P8	sector prediction	CKM/PMNS matrix non-primitivity	Matrix entries, phase conventions, angles, and basis choices are transition-coordinate skins or measurement/transport statuses.	A matrix entry or basis convention is required as primitive transition identity.
P9	sector prediction	Generation cardinality burden	A fourth generation is burden, failure, or target update unless role-cardinality closure is supplied.	A same-scope primitive generation addition preserves carrier closure without target update.
P10	sector prediction	New scalar/vector/fermion claims	New particle labels classify as representative, enrichment, route, projection, update, burden, or failure.	A new field label alone supplies primitive same-scope standing.

16.6 Falsification protocol

A serious falsifier must preserve the same target while supplying primitive standing outside the carrier network. A mere new observation, model, field label, parameter, or anomaly is not enough.

Step	Requirement
F1	Fix D_{SM} and the role bundle allegedly preserved.
F2	Exclude target update by showing that carrier inventory, invariant bundle, source state, admissibility boundary, and regime are unchanged.
F3	Exclude derivative statuses: representative, measurement, transport, route, mixing, projection, extension, and skin novelty.
F4	Supply primitive standing by providing the standing-bearing carrier or showing why a new primitive is required under fixed D_{SM} .
F5	Defeat descent to C_{SM} without destroying the claim.
F6	Close the certificate rather than leaving an unresolved counterexample burden.

Formally,

$$Falsifier(y) \Rightarrow \text{SameScope}_{D_{SM}}(y) \wedge \text{Primitive}(y) \wedge y \notin C_{SM} \\ \wedge \text{NonDerivative}(y) \wedge \text{ClosedCertificate}(y).$$

16.7 Evidence severity tiers

Tier	Meaning	Endpoint effect
Tier 0	Naming, model, label, or parameterization novelty.	No endpoint pressure unless carrier standing is supplied.
Tier 1	Statistically significant measurement anomaly.	Classify as witness, route stress, projection issue, transport correction, burden, or update candidate.
Tier 2	Repeated phenomenological pattern across routes or experiments.	Raises burden severity; may motivate extension disposition or target-update test.
Tier 3	Carrier-pressure evidence: anomaly cannot be represented as measurement, route, projection, or transport skin without losing standing.	Serious target-update or primitive-survivor candidate.
Tier 4	Same-scope primitive survivor: fixed D_{SM} preserved and independent primitive standing outside C_{SM} closes.	Falsifies same-scope exhaustion closure.

16.8 Falsifiable register format

Every prediction entry is required to specify claim type, expected status, excluded status, falsifier, evidence route, sector dependency, and current status. A prediction without an excluded status and falsifier is not a prediction-register entry; it is only a policy preference.

ID	Claim type	Expected status	Excluded status	Falsifier	Current status
P0	any future SM-associated claim	one of the fixed classifier statuses	ad hoc unregistered status	real claim cannot be classified without grammar expansion	closed as grammar commitment; falsifiable by future case
P1	BSM addition preserving D_{SM}	descent, derivative, burden, or failure	primitive standing by field/symmetry/-sector label alone	same-scope primitive BSM carrier outside C_{SM} with closed certificate	sector-relative / SM-6 target

P2	new observation or anomaly	measurement, route, projection, transport, burden, or update	primitive standing from observation alone	observation requires primitive carrier while preserving D_{SM} and defeating descent	classifier theorem plus sector evidence
P3	target-changing discovery	target update	same-scope survivor without fixed target	target-changing claim is validly counted as same-scope survivor	SM-6 synthesis target
P4	Higgs deviation or extension	measurement, route, projection, extension, burden, update, or failure	new primitive Higgs ground by scalar/vev/potential alone	same-scope primitive Higgs carrier outside $H_{\text{br/fix}}$	Higgs-arc supported; SM-6 received
P5	free quark claim	confinement-compatible matter standing or burden	isolated quark primitive by label alone	free quark with same-scope primitive SM standing	SM-3 target
P6	free gluon claim	non-asymptotic color transport or burden	free-gluon primitive by perturbative label alone	free gluon with same-scope primitive SM standing	SM-2/SM-3 target
P7	coupling/RG crossing	transport/calibration ledger	primitive standing from running value or plot	RG crossing generates carrier standing by itself	SM-4 target
P8	CKM/PMNS entry, angle, phase	mixing/transition coordinate, measurement, or transport status	primitive transition identity by matrix entry alone	basis-independent matrix coordinate required as primitive identity	SM-5 target
P9	fourth generation	role-cardinality burden, failure, or target update	field-duplication primitive family	same-scope fourth generation preserving carrier closure and role cardinality	high-risk SM-5 target
P10	new scalar/vector/fermion label	representative, route, projection, extension, burden, update, or failure	primitive standing by new field label alone	new label alone supplies primitive same-scope standing	sector-relative / BSM audit

17 Carrier-Identification Discipline

17.1 Carrier cannot be a favorite skin

A carrier is not a renamed preferred description:

$$\text{Carrier}(x) \neq \text{PreferredDescription}(x).$$

A carrier must be invariant under lawful redescription and must license a family of descriptions. If the purported carrier collapses when a gauge, basis, scheme, route, or projection is changed, it is not the carrier.

17.2 Carrier criteria

Criterion	Required question
Fixed target	What same-scope domain D_{SM} is being held fixed?
Role bundle	Which load-bearing role is preserved: gauge order, charge response, color transport, weak transition, matter representation, flavor transition, mass response, or projection interface?
Licensing relation	Which field symbols, particles, parameters, measurements, routes, and projections are licensed by the carrier?
Redescription stability	Does the carrier survive lawful changes of notation, gauge, basis, scheme, projection, or measurement route?
No hidden selector	Does the carrier secretly import a selector from a derivative skin?
No-promotion deletion	Which false primitive promotion is being deleted?
Burden	What same-scope survivor would have to be proven to defeat the classification?

17.3 Candidate versus closed carriers

SM-1 introduces the carrier network as a candidate audit map. A row becomes locally closed only when the relevant sector paper supplies the proof. SM-6 assembles the final carrier network and proves the residual primitive deletion theorem. This discipline protects SM-1 from premature sector conclusions.

18 Candidate Standard Model Carrier Register

Table 4: Candidate Standard Model carrier register v1

Carrier class	Candidate primitive role	Derivative skins / prediction pressure	Status
$C_{\text{gauge-order}}$	Local gauge-order standing and admissible interaction-structure preservation.	Gauge potential notation, gauge fixing, generator charts, propagators. Prediction: notation/gauge choice never self-promotes.	SM-2 target; upstream structural support.
$C_{\text{charge-response}}$	Unbroken or representation-indexed charge-response standing.	Charge numbers, currents, Coulomb display, detector signatures. Prediction: charge value is selector skin, not primitive object.	SM-4 target.
$C_{\text{color-transport}}$	Non-Abelian color-transport standing under confinement-compatible admissibility.	Color labels, perturbative gluons, jets, flux tubes. Prediction: no isolated primitive gluon/quark skin under fixed D_{SM} .	SM-2/SM-3 high-value closure.

Carrier class	Candidate primitive role	Derivative skins / prediction pressure	Status
$C_{weak-chiral}$	Weak chiral transition and charged/neutral response standing.	W/Z notation, weak mixing angle, pole masses, widths, current labels. Prediction: weak boson identity is carrier-resolved after electroweak split.	SM-2 target.
$C_{matter-rep}$	Representation-indexed matter-response standing.	Spinor fields, generation labels, masses, flavor names, tracks. Prediction: mass/flavor labels do not self-ground matter identity.	SM-3 target.
C_H	Higgs bridge/fixation carrier mediating order into deformation and mass-response standing.	Scalar, vev, potential, pole mass, vacuum, hierarchy, extension language. Prediction: Higgs deviations do not automatically create a new primitive Higgs ground.	Higgs arc supported; SM-2/SM-6 received.
$C_{flavor-transition}$	Admissible flavor-transition standing across bases, phases, and measured routes.	CKM/PMNS entries, angles, phases, oscillation baselines, CP displays. Prediction: matrix entries are coordinate skins.	SM-5 target; proof-risk.
$C_{projection-interface}$	Downstream metric, EFT, detector, and low-energy projection of carrier-indexed structure.	Planck ratios, EFT coefficients, event topologies, low-energy displays. Prediction: projection novelty is not primitive novelty.	SM-6 assembly target.

18.1 Proof-risk map

The high-risk rows are not hidden. Generation cardinality, fourth-generation claims, sterile-neutrino claims, extra scalars, quark/gluon isolated-standing claims, coupling/RG unification claims, and flavor matrix ontology require careful sector work. SM-1 records these rows as proof targets or burden loci rather than final dogma.

19 Worked Classification Examples

19.1 Higgs sector

The Higgs example is the proof of concept. The carrier is $C_H = H_{br/fix}$, the Higgs bridge/fixation carrier. Conventional scalar notation, vev language, potential terms, pole mass, hierarchy statements,

vacuum branches, and extension language are valid only after they receive their licensed statuses.

Submitted claim	Classifier result	Reason
The Higgs is primitively a scalar field.	LicensedRepresentative / FailedPrimitive if promoted.	Scalar notation is a representative of the bridge/fixation carrier, not the carrier itself.
The vev is the primitive Higgs ground.	Transport/measurement/scale skin; failed primitive if promoted.	A vev-like value is not a carrier; it is a display inside the electroweak route and scale ledger.
The potential is primitive Higgs identity.	Route/status skin.	Potential language can be lawful, but it is not the standing-bearing bridge/fixation role.
A second scalar automatically creates a new primitive Higgs.	ExtensionDisposition / Burden / TargetUpdate / FailedPrimitive.	Extra field labels do not self-ground a primitive. Same-scope preservation must be shown.
A lower vacuum branch is a primitive Higgs survivor.	Vacuum-route status, burden, target update, or failure.	Lower-branch availability is not primitive standing without same-scope carrier closure.

The Higgs arc therefore demonstrates the general rule: valid skins remain valid, but their primitive promotion is deleted [55, 56, 57, 58, 60, 61, 59].

19.2 Photon

A photon claim typically arrives through field quantum language, electromagnetic detector response, polarization, or scattering route. The classifier treats field quantum notation as representative. The primitive role, if locally closed, is unbroken electromagnetic charge-response and gauge-order standing. Detector response and polarization are measurement/route/projection statuses. A future photon-sector anomaly would be a measurement, route, projection, transport, burden, or target-update claim, not automatically a new primitive photon.

Standard gauge-theory usage already distinguishes presentation-dependent potentials and field representatives from gauge-invariant physical content. SM-1 uses that familiar orientation but reclassifies it as carrier standing versus licensed representation; the proof burden remains the carrier certificate, not ordinary gauge notation.

19.3 Gluon

The claim “a gluon is a free primitive particle like a photon” fails under confinement-compatible color-transport discipline unless a same-scope primitive free-gluon certificate closes. Perturbative gluon labels, color-flow diagrams, and jets are lawful route/projection statuses. The candidate primitive role is non-asymptotic color-transport standing. A free gluon with same-scope primitive standing would be a severe falsifier, not a mere naming novelty.

19.4 Quark

Quark field notation is a licensed representative of matter/color standing. Hadron membership, parton distributions, tracks, jets, and scattering signatures are route/projection statuses. The candidate primitive role is color-bearing matter standing under confinement-compatible target discipline. A

free isolated quark would matter only if it preserved D_{SM} , defeated derivative statuses, and supplied primitive standing under fixed carrier inventory.

19.5 Couplings and RG

A coupling value, running curve, matching condition, unification plot, or bare parameter is a transport/-calibration ledger unless a separate carrier certificate is supplied. RG crossings can be suggestive. They do not generate primitive standing alone. This distinction is essential for GUT and unification claims.

19.6 CKM and PMNS

CKM and PMNS entries, angles, phases, and basis choices are transition-coordinate skins unless a carrier-indexed transition certificate closes. Oscillation baselines and CP displays are route and measurement statuses. The candidate primitive role is flavor-transition standing across bases, phases, and measured routes.

In standard flavor usage, matrix coordinates depend on basis, phase convention, and parameterization even though invariant flavor phenomena are empirically meaningful. SM-1 preserves the empirical use of CKM/PMNS language while denying primitive transition identity by matrix entry alone.

19.7 Extra scalar

A new scalar field label is not a new primitive Higgs ground. The classifier asks whether the claim descends to C_H , preserves carrier inventory as enrichment, changes the target, remains burdened, or fails primitive promotion. Same-scope extra-scalar standing outside C_H would require a closed certificate.

19.8 Fourth generation

A fourth generation cannot be admitted by field duplication alone. It must preserve the representation-family ledger, anomaly/vacuum structure, Yukawa-realizability, mixing-coordinate dimension, and boundary bookkeeping without target update. Until that role-cardinality certificate closes, the claim is burden, failed primitive duplication, or target update.

19.9 Sterile neutrino

A sterile-neutrino claim is not one status. If weak-flavor-linked through a route or measurement, it may classify as route/status or extension disposition. If it introduces independent carrier inventory, it may be target update. If the same-scope carrier relation is ambiguous, it remains burden. The label “sterile neutrino” does not settle the primitive question.

19.10 Dark photon

A dark-photon claim may be projection/interface enrichment, representative extension, hidden-sector target update, or burden. A primitive electromagnetic carrier outside charge-response standing under fixed D_{SM} would require a severe certificate. Field-name similarity to the photon is not primitive continuity.

20 BSM and EFT Audit Matrix

20.1 Audit-admissible, not protected

BSM language is admissible for audit. It is not protected from same-scope primitive exhaustion.

EFT coefficients are powerful low-energy projection and matching tools. SM-1 classifies them as projection or transport statuses unless a coefficient is shown to require independent primitive standing under fixed D_{SM} . This does not deny the standard use of EFT operators, field redefinitions, or matching calculations. It separates their projection and transport authority from primitive carrier standing. The classifier does not ban BSM proposals; it asks what status they can carry under fixed D_{SM} , and whether their force comes from a skin, route, projection, target update, or genuine same-scope primitive certificate.

Claim type	Expected classifier result	Falsification pressure
Extra scalar	Carrier-preserving enrichment, role split, projection/route status, target update, burden, or failed primitive.	Same-scope primitive scalar standing outside C_H without target update.
Fourth generation	Role-cardinality burden, failed primitive duplication, or target update.	Same-scope primitive generation addition preserving fixed carrier closure.
Sterile neutrino	Route/status if weak-flavor-linked; target update if genuinely new carrier; burden if ambiguous.	Independent same-scope primitive neutrino standing outside matter/flavor network.
Dark photon	Projection/interface enrichment, representative extension, or target update.	Primitive electromagnetic carrier outside charge-response standing under fixed D_{SM} .
Supersymmetry	Target update unless it descends as derivative bookkeeping or carrier-preserving enrichment.	Same-scope primitive superpartner standing without changed carrier inventory.
Grand unification	Transport/unification skin or target update unless a carrier closure certificate is supplied.	Unification plot or RG crossing creates primitive standing alone.
Composite Higgs	Carrier-reconstruction claim or target update; not primitive by compositeness name alone.	Composite description displaces C_H while preserving D_{SM} and not changing carrier inventory.
EFT anomaly	Projection/transport/route stress, burden, or target-update candidate.	EFT coefficient by itself supplies primitive standing.

21 Dangerous BSM Certificates

The following certificates are included in the main text because they are the cases most likely to be pressed by a hostile referee. They show the classifier operating on serious BSM surfaces rather than only on favorable examples.

The BSM certificates are branch certificates, not single-model verdicts. SM-1 does not claim to decide every possible model instance in advance. It specifies the exact branch by which a submitted model becomes a licensed representative, extension disposition, burden state, target update, failed primitive, or genuine same-scope falsifier. A verdict range is therefore not indecision; it is the declared audit tree whose terminal branch is fixed by the submitted claim's certificate obligations.

Applying these certificates to named models requires the model’s own declared field content, symmetry assumptions, coupling structure, empirical claim surface, and target conditions. SM-1 supplies the branch logic, not a catalogue of model-by-model phenomenological verdicts.

The point of these certificates is not to adjudicate the phenomenological viability of each model class. A scalar extension, sterile-neutrino model, dark-photon model, EFT anomaly, supersymmetric model, grand-unified model, or composite-Higgs model may be useful, viable, or experimentally motivated in its own phenomenological literature. SM-1 asks the narrower primitive-status question: what carrier status, derivative status, burden, target update, or falsifier is actually being asserted relative to fixed D_{SM} ?

21.1 Extra scalar certificate

Field	Certificate entry
Submitted claim	A new scalar field supplies a new primitive Higgs ground.
Fixed target	D_{SM} with $C_H = H_{\text{br/fix}}$ and the Standard Model carrier network held fixed.
Same-scope test	The claim must preserve the Higgs bridge/fixation role, carrier inventory, invariant bundle, and admissibility boundary.
Skin test	Scalar notation, potential terms, mass eigenstate notation, mixing angle notation, and EFT scalar operators are skins unless a carrier certificate closes.
Carrier test	The claim must supply primitive standing outside $H_{\text{br/fix}}$ without merely renaming a scalar representative.
Target-update test	If the claim changes scalar-sector carrier inventory or admissibility boundary, it is a target update rather than a same-scope survivor.
Verdict range	Licensed representative, extension disposition, burden, target update, or failed primitive.
Falsifier	A same-scope scalar carrier with closed primitive standing outside $C_H = H_{\text{br/fix}}$, preserving D_{SM} and defeating derivative descent.
Sector status	Higgs-arc supported for $H_{\text{br/fix}}$; BSM/EFT closure received by SM-6.

21.2 Fourth generation certificate

Field	Certificate entry
Submitted claim	A fourth generation is simply another same-scope Standard Model generation.
Fixed target	D_{SM} with representation-family ledger, anomaly/vacuum survivor ledger, Yukawa-realizability topology, and mixing-coordinate dimension held fixed.
Same-scope test	The claim must preserve the family-cardinality envelope record rather than silently updating it.
Skin test	Field duplication, family label, additional mass values, and new mixing coordinates are not primitive standing by themselves.
Carrier test	The claim must supply role-cardinality closure showing how the added generation preserves carrier closure.
Target-update test	If the representation-family ledger or mixing-dimensionality changes, the claim updates the target.
Verdict range	Burden, failed primitive, or target update.
Falsifier	A same-scope fourth-generation carrier preserving the fixed carrier network and role-cardinality closure without target update.
Sector status	High-risk SM-5 target; not closed in SM-1.

21.3 Sterile neutrino certificate

Field	Certificate entry
Submitted claim	A sterile neutrino is a same-scope primitive neutrino.
Fixed target	D_{SM} with weak, matter-representation, and flavor-transition carrier roles held fixed.
Same-scope test	The claim must show whether it is weak/flavor linked inside the existing role bundle or introduces new carrier inventory.
Skin test	Sterile label, oscillation anomaly, mass state, cosmological bound, and model surface are not primitive identity by themselves.
Carrier test	The claim must supply independent primitive neutrino standing that does not descend to matter/flavor/route status.
Target-update test	A genuinely new hidden or sterile sector is a target update if it changes carrier inventory or admissibility boundary.
Verdict range	Route status, extension disposition, burden, target update, or failed primitive.
Falsifier	A sterile-neutrino primitive carrier that preserves D_{SM} , does not alter carrier inventory, and cannot descend to matter/flavor/route status.
Sector status	SM-3/SM-5 burden or target-update test; not closed in SM-1.

21.4 Dark photon certificate

Field	Certificate entry
Submitted claim	A dark photon is an additional primitive electromagnetic carrier.
Fixed target	D_{SM} with charge-response and projection-interface roles held fixed.
Same-scope test	The claim must preserve electromagnetic charge-response standing rather than adding a hidden-sector carrier.
Skin test	Dark-photon field label, kinetic-mixing notation, resonance surface, and detector topology are not primitive standing alone.
Carrier test	The claim must supply independent same-scope electromagnetic primitive standing outside charge-response carrier standing.
Target-update test	A hidden-sector carrier or new Abelian carrier inventory is target update unless same-scope preservation is certified.
Verdict range	Projection display, extension disposition, burden, target update, or failed primitive.
Falsifier	An independent same-scope electromagnetic primitive outside charge-response standing under fixed D_{SM} .
Sector status	SM-2/SM-4/SM-6 audit target.

21.5 EFT anomaly certificate

Field	Certificate entry
Submitted claim	A nonzero EFT coefficient or anomaly is primitive new physics inside the SM domain.
Empirical surface	Measurement anomaly, coefficient fit, threshold behavior, decay excess, or low-energy deviation.
Projection status	Operator basis, EFT coordinate, low-energy display, and matching condition are projection or transport statuses unless primitive standing is certified.
Carrier pressure	The anomaly becomes severe only if it cannot be represented as measurement, route, projection, or transport without losing standing.
Target-update test	If the anomaly requires changed invariant bundle, carrier inventory, or admissibility boundary, it is a target-update candidate.
Verdict range	Projection display, transport ledger, route status, burden, or target update.
Falsifier	A same-scope EFT anomaly whose standing cannot be represented as measurement, projection, route, or transport and which supplies a closed primitive carrier certificate outside C_{SM} .
Sector status	SM-4/SM-6 audit target; not closed in SM-1.

22 Source and Dependency Ledger for SM-1

22.1 Upstream kernel dependency and source direction

The dependency order is fixed as follows. The kernel/admissibility layer supplies the minimal non-degeneracy conditions for sameness, standing, reference, comparison, prediction, and falsification. SM-1 instantiates those conditions in the classifier arena; it does not derive the kernel from the Standard Model or from any later structural-spine paper. Later Standard Model structural results may instantiate the same kernel at larger scope, but they do not supply the upstream necessity of kernel preservation.

Accordingly, dependence on the kernel is not a defect in the proof class. A successful objection must either break a received local closure premise, supply a faithful same-scope primitive certificate, or recover non-degenerate physical claimhood without the kernel functions listed in Proposition 1.

22.2 Why the dependency ledger belongs inside the classifier paper

The master classifier is only audit-grade if the reader can see which upstream modules it uses and which claims it does not attempt to reprove. SM-1 therefore includes a dependency ledger. The ledger has three functions. First, it prevents the paper from silently importing the Standard Model as an unanalyzed premise. Second, it prevents the classifier from pretending to prove numerical parameter closure. Third, it fixes the denial obligations for a critic: the critic must attack a load-bearing module, the specialization from that module into D_{SM} , or the status assignment returned by the classifier.

The ledger is intentionally compressed. A support module may be cited here only for the output used by SM-1. The present paper does not need the full local proof of every upstream theorem. It needs the certified output: kernel legality, interface discipline, realized physical interior, gauge/product structural recovery, Higgs bridge/fixation status, Yukawa-realizability status, and the Higgs carrier-arc prototype for no primitive skin promotion.

22.3 Load-bearing module table

Table 5: SM-1 dependency ledger

Layer	Imported output	Use in SM-1	Status in this paper
AASC kernel	Admissibility, standing, reference, scope, bivalence, no-repair discipline.	Licenses the distinction between standing-bearing carriers and derivative descriptions.	received
Interface / AMetric	No primitive metric, coordinate, selector, or scale standing prior to admissible fixation.	Blocks field, mass, coordinate, vev, and metric display from primitive promotion.	received
Anchor–Tensor–Skin	Separation between standing anchor, load-bearing tensor, and display skin.	Supplies the general form of no-promotion: skin display cannot create standing.	received
Realized physical interior	Physical structure enters as boundary-licensed persistence and carrier organization.	Prevents the classifier from floating above physics as pure vocabulary.	received
Gauge realization / PSM / SR	Gauge-product, representation-response, charge, anomaly, vacuum, and residual survivor structure.	Supplies the structural Σ_{SM} payload and D_{SM} arena.	received
Standalone SM spine	Structural tuple and parameter/nonclaim boundary.	Defines what SM-1 may hold fixed without claiming full numerical closure.	received
Higgs bridge carrier arc	$C_H = H_{\text{br/fix}}$, no primitive scalar/vev/scale/vacuum/extension promotion.	Prototype for component normal form and worked examples.	received / exemplar
Higgs extension classifier	Extra scalar and same-scope preservation discipline.	BSM/EFT audit model for extra scalar claims.	exemplar
QueryPack / control matrix	Source pool, machinery vocabulary, dependency edges, supersession discipline.	Keeps terminology and dependency posture aligned with current corpus state.	operational

22.4 Availability and audit status of dependencies

For external audit, the dependency ledger distinguishes proof use from availability status. The present manuscript uses upstream modules only at the output status declared in the ledger; the availability column records how a reader can locate or audit the relevant source in the current project package or public archive.

Dependency family	Availability status	SM-1 proof use	Audit role
AASC kernel	Public Lean/archive where available; kernel Lean archive DOI: 10.5281/zenodo.20546437.	Load-bearing for SM-1.	Supplies admissibility, standing, reference, bivalence, and no-repair discipline.
Project plan for SM carrier closure	Companion project-plan source in the release package.	Staging and scope control.	Fixes the SM-1 through SM-6 staging, prediction register, and falsification protocol.
Standalone Standard Model proof spine	Companion PDF and source project in the release package.	Load-bearing for the received arena.	Supplies the structural tuple Σ_{SM} and the structural/parameter nonclaim boundary.
Higgs carrier arc	Companion Higgs theorem package.	Load-bearing exemplar; sector-specific closure received only for Higgs-prototype uses.	Supplies the prototype carrier/skin separation and $C_H = H_{\text{br/fix}}$.
AASC Query Core / QueryPack	Companion operational corpus package.	Operational, not independent proof.	Supplies vocabulary, dependency posture, and supersession discipline.
External standard-physics literature	Bibliographic orientation references.	Orientation only.	Fixes conventional usage being reclassified; does not supply the AASC proof.
Later sector papers SM-2–SM-6	Future project deliverables, not assumed closed in SM-1.	Not used as closed proof in SM-1.	Convert sector-relative schemas into theorems only when their local closures are supplied.

22.5 Dependency nonclaims

The dependency ledger also fixes what SM-1 does not inherit. It does not inherit a completed derivation of the full empirical parameter table. It does not inherit sector-level closure for every candidate carrier in the register. It does not inherit a proof that every possible BSM target is false. It inherits the structural arena and the status machinery needed to classify claims inside or relative to that arena.

Principle 22.1 (Dependency discipline). A downstream classifier may use an upstream module only at the output status certified by that module. A structural payload may not be silently promoted into a numerical parameter closure; a candidate carrier may not be silently promoted into final sector closure; a burden state may not be silently promoted into a survivor.

23 Classifier Stage Certificates

23.1 Why stage certificates are needed

The classifier must be reproducible. A reader should be able to take a submitted claim, run it through the same tests, and see why the result is carrier, representative, measurement, transport, route, mixing,

projection, extension, target update, burden, or failure. This section gives the stage-level certificates that make the classifier audit-grade.

23.2 Target-stage certificate

The target-stage certificate is

$$Cert_x^{target} = (D_x, Inventory_x, Inv_x, Boundary_x, Regime_x, RoleBundle_x, Result_x).$$

It asks whether the submitted claim preserves the Standard Model arena or changes it. A claim cannot move carrier inventory, add hidden load, change the admissibility boundary, or change the physical regime while remaining a same-scope primitive challenge.

Target-stage item	Audit question
Declared target	Is the claim explicitly evaluated under D_{SM} , or has it changed the target?
Carrier inventory	Does it preserve the carrier inventory, or add a new standing-bearing object?
Invariant bundle	Does it preserve charge, anomaly, vacuum, representation, and role constraints?
Admissibility boundary	Does it preserve the same boundary and regime conditions?
Source state	Does it preserve the source/measurement domain used by D_{SM} ?
No hidden load	Does it secretly introduce new degrees of freedom while claiming same-scope status?
Result	Same-scope, target update, or insufficiently specified.

23.3 Carrier-stage certificate

The carrier-stage certificate is

$$Cert_x^{carrier} = (Role_x, Invariant_x, Licensing_x, Redescription_x, NoSelector_x, Burden_x).$$

It is the central guard against vocabulary substitution. The submitted object must be shown to preserve a role under redescription and to license derivative skins. A skin cannot be declared a carrier merely because the author prefers it.

23.4 Descent-stage certificate

The descent-stage certificate is

$$Cert_x^{descent} = (Rep_x, Meas_x, Transport_x, Route_x, Mix_x, Proj_x, Extension_x, DescentResult_x).$$

It asks whether the claim can be placed downstream of an existing carrier. Most ordinary physics language survives here. The point of the classifier is not to delete field language, measurement language, or EFT language. The point is to place them in the right status.

Descent channel	Typical objects placed there
Representative	field notation, particle label, generator chart, conventional Lagrangian display
Measurement	pole mass, width, detector signature, cross-section deviation, oscillation datum
Transport	RG flow, scheme conversion, threshold matching, basis change, calibration relation
Route	decay channel, scattering path, confinement route, oscillation route, thermal history
Mixing	CKM/PMNS coordinate, phase convention, basis angle, transition display
Projection	EFT operator, detector topology, low-energy display, metric ratio, phenomenological plot
Extension	same-scope enrichment, role split, carrier-preserving BSM/EFT disposition

23.5 Prediction-stage certificate

The prediction-stage certificate is

$$Cert_x^{pred} = (PredictionID_x, ExpectedDisposition_x, NonPromotion_x, Falsifier_x, EvidenceRoute_x, CurrentStatus_x).$$

It prevents prediction claims from being post hoc. The classifier must state in advance what kind of future claim is governed, which disposition is expected, what cannot self-promote, and what would falsify the prediction.

23.6 Burden-stage certificate

The burden-stage certificate is

$$Cert_x^{burden} = (SameScope_x, NonDescent_x, PrimitiveAssertion_x, OutstandingProof_x, Severity_x).$$

A burden is not a failure of the framework. It is a controlled review state. It becomes dangerous only if the project counts it as a survivor without closure. SM-1 forbids that promotion.

23.7 Failure-stage certificate

The failure-stage certificate is

$$Cert_x^{fail} = (AssertedPrimitive_x, SuppliedObject_x, SkinType_x, MissingCarrier_x, FailureLine_x).$$

It is assigned when a primitive assertion supplies only a skin. The failure line should be exact, for example: “field label supplied, carrier role not supplied,” or “matrix entry supplied, transition carrier not supplied.”

24 Raw Role Grammar and Disposition Grammar

24.1 Raw role grammar

Before final disposition, the classifier may tag a claim with raw role descriptors. These descriptors do not by themselves settle primitive status; they identify the channel through which the claim is submitted.

Table 6: Raw role grammar

Raw role	Typical surface	Classifier danger
Field-surface	scalar, spinor, gauge potential, tensor notation	field notation mistaken for carrier standing
Particle-surface	photon, gluon, quark, neutrino, Higgs, new resonance	label mistaken for primitive identity
Value-surface	mass, charge, coupling, width, vev, self-coupling	number mistaken for standing source
Matrix-surface	CKM, PMNS, mass matrix, Yukawa texture	coordinate/basis skin mistaken for transition identity
Gauge-surface	gauge choice, generator chart, ghost, propagator	representational redundancy mistaken for ontology
Measurement-surface	detector event, pole mass, anomaly, decay width	witness mistaken for primitive generator
Route-surface	decay, scattering, confinement, oscillation, thermal path	route availability mistaken for primitive status
Projection-surface	EFT coefficient, low-energy operator, event topology	projection display mistaken for new primitive
Model-surface	BSM sector, symmetry, unification group, compositeness name	model vocabulary mistaken for same-scope carrier

24.2 Disposition grammar

Final disposition is stricter than raw role. It answers what the claim can be after the target, carrier, descent, prediction, burden, and failure stages.

Table 7: Disposition grammar

Disposition	Positive condition	Negative boundary
Carrier	role-bearing standing closes under fixed D_{SM}	cannot be merely a display or preferred notation
Licensed representative	display is licensed by a carrier	cannot self-promote into primitive standing
Measurement witness	empirical datum witnesses carrier response or route stress	cannot generate the carrier it witnesses
Transport ledger	comparison, matching, scheme, RG, calibration, or basis map preserves/routes standing	cannot create primitive standing by chart choice
Route status	process path or phenomenological route is downstream of carriers	cannot become primitive by route availability
Mixing status	matrix/phase/basis data record transition coordinates	cannot become primitive transition identity by entry value alone

Disposition	Positive condition	Negative boundary
Projection display	EFT, detector, metric, or low-energy surface displays carrier-indexed structure	cannot become primitive by projection success
Extension disposition	addition preserves or enriches carrier roles without same-scope primitive escape	cannot hide target update as same-scope addition
Target update	claim changes carrier inventory, regime, source state, boundary, or invariant bundle	cannot be counted as same-scope survivor
Counterexample burden	same-scope primitive survivor asserted but certificate open	cannot be treated as survivor
Failed primitive	primitive claim supplies only skin/model/route/value/label	cannot be salvaged without new carrier certificate

25 Sector Prediction Boxes

25.1 Gauge prediction box for SM-2

Gauge prediction item	Content
Claim class	Photon, gluon, W, Z, gauge potential, polarization, propagator, gauge fixing, pole mass, width, detector signature, jet, color-flow display.
Expected disposition	Carrier if the standing-bearing gauge/color/weak role is closed; otherwise representative, measurement, transport, route, projection, burden, target update, or failure.
Non-promotion rule	Field notation, gauge choice, polarization, propagator behavior, pole mass, width, detector event, jet signature, or color-flow display does not self-promote into primitive gauge standing.
Falsifier	A same-scope primitive gauge-boson standing outside the gauge/color/weak carrier network with no target update and no derivative descent.

25.2 Matter prediction box for SM-3

Matter prediction item	Content
Claim class	Quarks, charged leptons, neutrinos, mass labels, flavor labels, detector tracks, oscillation routes, cosmological bounds, current/constituent masses, hadron membership.
Expected disposition	Matter-representation carrier if role closes; otherwise representative, measurement, route, transport, projection, burden, target update, or failure.
Non-promotion rule	Mass, flavor label, detector track, oscillation route, cosmological bound, current mass, constituent mass, hadron membership, or isolated-particle skin does not self-promote into primitive matter identity.
Falsifier	A free quark or free gluon with same-scope primitive SM standing, or a mass/flavor label that supplies primitive matter identity without carrier descent.

25.3 Transport prediction box for SM-4

Transport prediction item	Content
Claim class	Bare charge, charge number, coupling value, running crossing, threshold matching, unification plot, scheme coordinate, calibration convention.
Expected disposition	Charge-response carrier or transport/calibration ledger, depending on role; not primitive from value alone.
Non-promotion rule	No bare charge number, coupling value, running crossing, matching condition, unification plot, or scheme coordinate generates primitive standing without carrier-normal-form descent.
Falsifier	An RG crossing or coupling value that generates primitive standing by itself under fixed D_{SM} .

25.4 Flavor prediction box for SM-5

Flavor prediction item	Content
Claim class	CKM, PMNS, CP phases, basis choices, fitted angles, generation labels, fourth-generation claims, flavor anomalies.
Expected disposition	Flavor-transition carrier if role closes; otherwise transition-coordinate skin, measurement witness, route status, burden, target update, or failure.
Non-promotion rule	No CKM/PMNS entry, basis choice, phase convention, fitted angle, or flavor anomaly functions as primitive transition identity by itself.
Falsifier	A matrix entry, basis convention, or same-scope fourth-generation addition that supplies primitive standing while preserving fixed carrier closure.

25.5 Synthesis prediction box for SM-6

SM-6 prediction item	Content
Claim class	Any future SM-associated claim after sector closures are received.
Expected disposition	Carrier, representative, measurement, transport, route, mixing, projection, extension, target update, burden, or failure.
Non-promotion rule	Future physics may add skins, routes, measurements, projections, and target updates; it cannot add unlicensed same-scope primitive standing inside closed D_{SM} .
Falsifier	A closed same-scope primitive survivor outside C_{SM} that is not target update, derivative status, burden state, or failed primitive.

26 Detailed Classification Certificates for Core Examples

26.1 Higgs certificate

Certificate field	Higgs classification
Submitted object	Scalar field, vev, potential, pole mass, lower vacuum branch, extra scalar, hierarchy statement, extension claim.

Fixed target	Standard Model structural arena with $H_{\text{br/fix}}$ as Higgs bridge/fixation carrier.
Candidate carrier	$C_H = H_{\text{br/fix}}$, the bridge/fixation carrier mediating admitted order into deformation and mass-response standing.
Licensed representatives	Conventional scalar doublet notation, field excitation language, scalar potential display, vev-like coordinate, pole mass measurement, weak-scale route language.
Derivative skins	Scalar coordinate, vev, potential coefficient, normalization, empirical scale, mass matrix, mixing datum, lower-branch display, extra-scalar model name.
Prediction status	P4 Higgs-sector deviations; P10 new scalar/vector/fermion claims.
False primitive deleted	No scalar, vev, potential, scale, lower branch, or extra-field label supplies primitive Higgs standing by itself.
Falsifier	A same-scope Higgs primitive outside $H_{\text{br/fix}}$ preserving D_{SM} , defeating descent, excluding target update, and closing primitive standing.
Endpoint line	Higgs arc supports carrier occupation and no fifth identity ground; SM-1 imports it as prototype and handoff.

26.2 Photon certificate

Certificate field	Photon classification
Submitted object	Photon field quantum, electromagnetic wave, polarization state, detector response, scattering route.
Fixed target	Gauge/charge-response sector under D_{SM} .
Candidate carrier	Unbroken electromagnetic charge-response and gauge-order standing.
Licensed representatives	Gauge field notation, photon label, polarization basis, propagator display, S-matrix route language.
Derivative skins	Gauge choice, potential notation, detector event, polarization coordinate, field quantum label.
Prediction status	P0 universal sorting; P10 new vector claims; SM-2 gauge prediction box.
False primitive deleted	Field quantum notation and detector response do not self-promote into primitive photon standing.
Falsifier	Same-scope electromagnetic primitive outside charge-response/gauge-order carrier with no derivative descent.
Endpoint line	SM-2 must close the gauge-sector carrier theorem.

26.3 Gluon certificate

Certificate field	Gluon classification
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Submitted object	Perturbative gluon, jet, color-flow diagram, free-gluon claim, confinement route.
Fixed target	QCD color-transport sector under confinement-compatible D_{SM} .
Candidate carrier	Non-Abelian color-transport standing.
Licensed representatives	Gluon field notation, perturbative internal/external labels where admissible, color-flow diagrams, jet route language.
Derivative skins	Free-particle picture, jet topology, color label, diagrammatic line, perturbative display.
Prediction status	P6 gluon non-asymptotic standing.
False primitive deleted	Perturbative label, jet, or color-flow display does not supply isolated primitive gluon standing.
Falsifier	A free gluon with same-scope primitive SM standing under fixed D_{SM} .
Endpoint line	SM-2/SM-3 must close non-asymptotic color-transport status.

26.4 Quark certificate

Certificate field	Quark classification
Submitted object	Quark field, flavor label, current mass, constituent mass, parton route, hadron membership, isolated-quark claim.
Fixed target	Matter/color sector under confinement-compatible D_{SM} .
Candidate carrier	Color-bearing matter standing with representation-indexed response.
Licensed representatives	Spinor field notation, flavor label, current/constituent mass displays, hadron route language, parton distribution displays.
Derivative skins	Isolated-particle picture, mass label, flavor name, detector route, hadron membership.
Prediction status	P5 quark confinement standing.
False primitive deleted	Isolated-particle skin and mass/flavor labels do not supply primitive quark identity.
Falsifier	A free quark with same-scope primitive SM standing preserving D_{SM} .
Endpoint line	SM-3 must close matter carrier status.

26.5 Coupling/RG certificate

Certificate field	Coupling/RG classification
Submitted object	Bare coupling, measured coupling, running coupling, RG crossing, matching threshold, unification plot.
Fixed target	Charge/coupling/transport sector under D_{SM} .
Candidate carrier	Charge-response and interaction-strength transport/calibration standing.
Licensed representatives	Coupling values, scheme coordinates, beta-function displays, matching equations, unification plots.

Derivative skins	Bare value, scheme coordinate, plotted crossing, normalization convention, calibration value.
Prediction status	P7 coupling/RG transport.
False primitive deleted	No coupling value, running crossing, matching condition, or unification plot supplies primitive standing alone.
Falsifier	An RG crossing or bare coupling value that generates primitive standing without carrier normal form.
Endpoint line	SM-4 must close transport non-generation and charge/coupling ledger status.

26.6 CKM/PMNS certificate

Certificate field	Flavor matrix classification
Submitted object	CKM entry, PMNS entry, mixing angle, CP phase, basis choice, fitted anomaly, oscillation route.
Fixed target	Flavor-transition sector under D_{SM} .
Candidate carrier	Admissible flavor-transition standing across bases, phases, and measured routes.
Licensed representatives	Matrix parametrizations, angles, phases, oscillation baselines, CP displays, fitted transition coordinates.
Derivative skins	Matrix entry, phase convention, flavor basis, parametrization choice, baseline route.
Prediction status	P8 matrix non-primitivity; P9 generation cardinality burden where cardinality is implicated.
False primitive deleted	No matrix entry, basis choice, phase convention, or fitted angle supplies primitive transition identity by itself.
Falsifier	A matrix entry or basis convention required as primitive transition identity under fixed D_{SM} .
Endpoint line	SM-5 must close flavor-transition carrier status and generation burden.

27 Denial Obligations and Falsifier Checklist

27.1 Denial obligations

SM-1 denies primitive promotion only under a declared burden. The denial obligation associated with a false primitive is not “show that the skin is useless.” The obligation is “show that the skin lacks independent primitive standing under fixed D_{SM} .” This is a narrower and more defensible denial.

Table 14: Denial obligation register

False primitive	What SM-1 denies	What SM-1 does not deny
Field symbol	Primitive standing by notation alone.	Use of fields as lawful representatives and calculational objects.
Particle label	Primitive identity by name alone.	Use of particle labels in ordinary phenomenology.
Mass value	Primitive carrier status by measured value.	Measurement, pole/running discipline, ratio or parameter-sector programs.
Coupling value	Primitive standing by bare/running value.	Couplings as transport/calibration ledgers and possible downstream targets.
Matrix entry	Primitive transition identity by coordinate entry.	Matrices as transition-coordinate displays and measurement summaries.
Gauge choice	Ontic multiplicity or primitive object by gauge presentation.	Gauge methods as admissible redescription machinery.
Measurement anomaly	Automatic new primitive.	Serious evidence pressure, burden escalation, or target-update motivation.
EFT coefficient	Primitive standing by projection coefficient.	EFT as projection, matching, route, or diagnostic language.
BSM model name	Protected primitive alternative by naming.	Audit admissibility and target-updating discoveries.

27.2 Falsifier checklist

A proposed falsifier must pass all of the following. Failure at any step does not make the claim uninteresting; it changes its status.

1. It must be a real submitted claim with a specified empirical or formal surface.
2. It must declare the fixed D_{SM} it allegedly preserves.
3. It must show that carrier inventory, invariant bundle, source state, admissibility boundary, role bundle, and regime are unchanged.
4. It must show that it is not a representative, measurement witness, transport ledger, route status, mixing status, projection display, or extension disposition.
5. It must supply a primitive standing certificate rather than merely a new label or model surface.
6. It must defeat descent to the registered carrier network.
7. It must close rather than remain a counterexample burden.

28 Manuscript Architecture for the Full SM Sequence

28.1 SM-1 deliverables

SM-1 delivers the master classifier, the normal form, the no-promotion theorem family, target-update separation, future-claim classifiability, a candidate carrier register, the prediction register, falsification protocol, and worked certificates. It does not deliver final sector closure.

28.2 Sector-paper dependency order

The dependency order is gauge, matter, charges/couplings/RG, flavor/mixing/generation, then synthesis. This order is not arbitrary. Gauge and matter define many of the carrier roles that charge/coupling and flavor/mixing rely upon. Flavor/generation has the highest proof risk and should not be used as a premise for earlier sections.

Paper	Primary endpoint	SM-1 handoff
SM-2	Gauge carriers: photon, gluon, W, Z.	Gauge prediction box, no field/gauge/polarization/pole-mass primitive promotion.
SM-3	Matter carriers: quarks, leptons, neutrinos.	Matter prediction box, no mass/flavor/track primitive promotion, confinement-compatible burden.
SM-4	Charges, couplings, and renormalization as standing transport.	Transport prediction box, no primitive standing from bare coupling or RG crossing.
SM-5	Flavor transition, mixing, and generation cardinality.	Flavor prediction box, CKM/PMNS non-primitivity, generation role-cardinality burden.
SM-6	Standard Model carrier exhaustion closure.	Receives sector closures, assembles C_{SM} , proves residual primitive deletion.

29 Minimal Hostile Reconstruction

29.1 Kernel-preserved hostile reconstruction

A hostile reconstruction has force only when it is a same-target reconstruction. It must preserve D_{SM} , preserve the comparison class, preserve reference standing, and then supply an independent primitive role not already carried or licensed by the closed classifier package. Merely replacing the target, changing the admissibility rule, or appealing to a useful model surface is not a same-scope counterexample.

Criterion 29.1 (Faithful classifier counterexample). A faithful classifier counterexample must show all of the following: (i) kernel-preserved same-scope membership in D_{SM} ; (ii) primitive role occupancy; (iii) non-derivativeness from the certified classifier carriers; (iv) nonmembership in the registered status atlas; and (v) target-preserving failure of the no-promotion theorem. If any item fails, the claim is a registered status, burden, failed primitive, or target update rather than a primitive survivor.

A serious audit must formulate the strongest possible objection in the opponent's terms. Let y be a future Standard Model-associated claim satisfying all of the following:

1. $\text{FutureSMClaim}(y)$;

2. $\text{SameScope}_{D_{\text{SM}}}(y)$;
3. $y \notin C_{\text{SM}}$;
4. y is not a field label, particle name, measurement witness, route, projection, EFT coordinate, matrix coordinate, basis convention, scheme coordinate, model name, or transport ledger;
5. y does not change carrier inventory, invariant bundle, boundary, source state, role bundle, or comparison arena;
6. y supplies a closed primitive carrier certificate outside C_{SM} .

Then

$$\text{Survprim}_{D_{\text{SM}}}(y \setminus C_{\text{SM}})$$

would hold. This would defeat the same-scope primitive-exhaustion endpoint. SM-1 does not reclassify such a claim as target update, burden, or failed primitive once its certificate closes. The classifier is falsifiable precisely because this case is admitted as a possible defeat condition.

The live dispute is therefore not whether a hostile counterexample is conceivable. It is whether any concrete Standard Model-associated claim can satisfy all six obligations while preserving the fixed target and defeating the carrier/skin/derivative descent tests.

30 Referee Objection Lock

30.1 “This denies new physics”

It does not. It denies only unlicensed same-scope primitive promotion. New observations, anomalies, EFT displays, routes, projections, and target updates remain possible. The classifier makes them more precise.

30.2 “This is just metaphysics”

The paper is a status theorem over the Standard Model comparison arena. It classifies physical claims by carrier standing, derivative status, target update, burden, or failure. It is not a phenomenological replacement model and does not compete with QFT calculations at the level of cross sections or amplitudes.

30.3 “The Standard Model already defines particles by fields”

Field representation is granted. Primitive standing by field notation alone is denied. This is the same distinction already needed in gauge theory, confinement, flavor bases, scheme dependence, and the Higgs bridge/fixation arc.

30.4 “Predictions are not numerical”

The predictions are primitive-status predictions with stated falsifiers. Not all testability is numerical first-order prediction. A same-scope primitive survivor outside the carrier network would falsify the relevant prediction. A real claim requiring ad hoc classifier expansion would falsify P0 and the non-ad-hoc classifiability commitment.

30.5 “Carrier candidates are speculative”

SM-1 marks the network as a candidate register. The sector papers must close local carriers. SM-6 assembles the final endpoint. Premature assertion is avoided by design.

30.6 “AASC dependency is a defect”

AASC dependence is the proof class. The question is whether the dependency is declared and whether the local uses are typed correctly. A same-target objection must break the dependency, invalidate the specialization, or provide a faithful same-scope counterexample preserving the fixed target and carrier package.

31 Lean 4 formalization target and public audit reference

The Lean 4 release is used here as a paper-shaped, proof-checked audit and certificate surface for the SM manuscript sequence. It records the endpoint carriers, declared hypotheses, theorem crosswalk inventory, and certificate interfaces in Lean; verifies that the stated closure theorems typecheck without `sorry`, `admit`, live local `axiom` declarations, or `unsafe` code in the audit surface; and ties the formal release to the DOI-bearing manuscript bundle. This is a substantive support layer: it gives a typed theorem registry, explicit hypothesis boundaries, machine-checked closure over declared certificate fields, an audit trail across SM-1–SM-6 and the endpoint paper, reproducibility checks, and DOI/release alignment. It should not be described as an independent reconstruction of the full Standard Model argument from foundational physics.

The generation-cardinality endpoint should be read at the governance-to-endpoint interface, not as a standalone phenomenological derivation inside the Standard Model sector. The mechanized surface represents the pre-physical admissibility constraints under which a non-degenerate physical presentation is admissible: open-arena discipline, no-selector and no-presentation-dependence constraints, role-occupancy closure, and boundary handoff. Within that framework, generation cardinality is a formal closure condition rather than a fit to particle content or an inference from arbitrary presentation-dependent structure. The Lean release audits that certificate surface and its dependencies; it does not reconstruct every analytic or phenomenological argument in the SM papers.

The public Lean-facing support package is the relevant SM ARC reference archive [73]: its README identifies the paper-shaped theorem surface, the locally received SM-1–SM-6 ARC layers, the generation-cardinality endpoint, the theorem-inventory layer, and the build/audit verifier. Its v0.1.0 release records the `GenerationCardinality` Lake target, carries SM-1 through SM-6 as local received ARC layers, preserves the generation-cardinality source artifacts, and reports build and focused axiom-audit checks over the endpoint anchors.

Lean audit item	Manuscript-facing role
Repository	somamaley-ux/AASC-Standard-Model-Carrier-Exhaustion-Generation-Cardinality, the public Lean 4 reference archive for the SM-1–SM-6 ARC carrier-exhaustion chain and the generation-cardinality endpoint.
Release	v0.1.0 - Standard Model Carrier Exhaustion / Generation Cardinality.

DOI	10.5281/zenodo.21219417.
Primary theorem surface	<code>SM1.lean</code> and its imported <code>SM1/...</code> files, together with <code>SM1/TheoremInventory.lean</code> where present; these are read against <code>SM2.lean</code> – <code>SM6.lean</code> and <code>GenerationCardinality.lean</code> as downstream support surfaces.
Axiom audit command	Repository verifier <code>scripts/check-generation-cardinality-audit.ps1</code> with the <code>-SkipLakeUpdate</code> option; the verifier builds the endpoint target, scans for live <code>axiom</code> , <code>sorry</code> , <code>admit</code> , and <code>unsafe</code> declarations, and runs focused axiom checks over endpoint anchors.
Build target	<code>GenerationCardinality</code> , with aggregate imports through <code>SM1.lean</code> – <code>SM6.lean</code> and <code>GenerationCardinality.lean</code> .
Audit meaning	The public release machine-checks the governance-to-endpoint certificate surface over declared carriers, received ARC layers, and explicit manuscript hypotheses. Once the pre-physical admissibility layer is accepted, the endpoint predicate is a formal closure condition, not a phenomenological fit or downstream presentation selector.
Certificate/interface value	The layer is not merely bookkeeping: it supplies a typed theorem registry, explicit hypothesis boundaries, machine-checked closure over declared certificate fields, an SM-1–SM-6 audit crosswalk, reproducibility checks, and DOI/release alignment.
Boundary	The cardinality endpoint is not a standalone SM-sector derivation from particle dynamics; it is a governance-level endpoint that makes explicit the admissibility, no-selector, open-arena, role-occupancy, and presentation-independence constraints under which later SM interpretations are meaningful. The Lean release audits this certificate/interface boundary and its dependencies; it does not reconstruct every analytic or phenomenological argument of SM-1–SM-6.

A manuscript-faithful Lean route for SM-1 represents:

1. fixed scopes and submitted-claim or candidate-record sets as explicit carriers;
2. carrier packages, registered-status packages, and received sector packages as typed structures rather than implicit background metaphysics;
3. same-scope and target-update predicates as explicit boundary conditions;
4. derivative-skin, representative, route, projection, transport, burden, failure, and extension statuses as typed predicates or classifiers;
5. certificate closure and no-promotion principles as theorems over declared hypotheses, not as global axioms;
6. finite branch exhaustions for the relevant same-scope arena;
7. the handoff to SM-6, or from SM-6 to the generation-cardinality endpoint, as an explicit typed support relation.

For SM-1, the paper-specific audit surface is the master classifier: claim submission, same-scope tests, target-update separation, derivative-skin non-promotion, carrier-certificate discipline, prediction-register typing, counterexample burden, failed primitive status, and the sector-relative residual primitive deletion

schema. These are not future desiderata; they are the status predicates and theorem anchors against which the public repository is read.

A manuscript-faithful formalization over the declared carriers and hypotheses requires no custom axiom at a load-bearing node. If a proposed audit file relies on **axiom**, **sorry**, **admit**, **unsafe**, or an opaque proof-carrying placeholder at such a node, it is not counted as formal support for this manuscript. Conversely, a clean axiom audit is not a stand-alone validation of the physical hypotheses encoded as record fields; it is a proof-checked governance-to-endpoint certificate boundary whose value is typed traceability, explicit assumptions, reproducible build discipline, and DOI-aligned auditability.

32 Handoff to Sector Papers

32.1 SM-2: Gauge carriers

SM-1 supplies the gauge-order carrier definition, no primitive field/gauge/polarization/pole-mass promotion, and the prediction box for photon, gluon, W, and Z claims. SM-2 must prove local gauge-sector carrier closure.

32.2 SM-3: Matter carriers

SM-1 supplies the matter-representation carrier definition, no mass/flavor/track primitive promotion, quark/gluon confinement-standing burdens, and neutrino route/status typing. SM-3 must prove local matter-sector carrier closure.

32.3 SM-4: Charges, couplings, and RG

SM-1 supplies charge-response carrier language, transport-ledger status for couplings/RG, and no primitive standing from bare coupling or running crossing. SM-4 must close the charge/coupling/RG transport theorem.

32.4 SM-5: Flavor, mixing, and generation

SM-1 supplies flavor-transition carrier language, CKM/PMNS matrix non-primitivity, and role-cardinality burden for fourth-generation claims. SM-5 must close flavor/mixing/generation or record a controlled burden.

32.5 SM-6: Final endpoint

SM-6 receives the master classifier, sector closures, prediction boxes, and BSM audit machinery. It proves the final endpoint:

$$\text{Survprim}_{D_{\text{SM}}}(\text{Attr}_{\text{SM}} \setminus C_{\text{SM}}) = 0.$$

33 Conclusion

The kernel-status strengthening is part of the theorem, not a commentary layer. The closure claimed here is primitive-status closure under the minimal non-degeneracy conditions required for classifier claims to remain the same standing-bearing physical claims across measurement, prediction, comparison, and attempted falsification.

SM-1 does not close the Standard Model empirically. It closes the classifier-level primitive-promotion route for field symbols, particle labels, measured values, couplings, matrices, gauges, routes, projections, EFT displays, and model names when they satisfy the skin-identification criteria and lack an independent carrier certificate. Future physics remains possible as measurement, route, projection, transport, extension, burden, failure, or target update. What is denied is unlicensed same-scope primitive escape by display, measurement, route, or model name alone. The later sector papers test the carrier register locally; SM-6 assembles the final exhaustion closure.

A Component Audit Template

Field	Audit question
Submitted object	What object, label, field, parameter, matrix, route, or measurement is being evaluated?
Fixed target	What same-scope target D is active?
Candidate carrier	What standing-bearing role must be occupied for admissibility?
Role bundle	Which functions must be preserved: gauge inheritance, charge response, color transport, weak transition, matter representation, flavor transition, mass response, projection interface?
Derivative skins	Which field symbols, labels, masses, charges, couplings, measurements, gauges, routes, or projections are lawful but nonprimitive?
Route statuses	Which decay, scattering, confinement, oscillation, lifetime, thermal, detector, or phenomenological routes are downstream?
Transport ledgers	Which RG, matching, scheme, basis, calibration, or projection maps are required?
Target updates	Which changes add carrier inventory, source state, EFT threshold, BSM sector, compositeness, gravitational rule, or boundary change?
Prediction status	Which prediction-register item applies?
False primitive	Which primitive promotion is being deleted?
Counterexample burden	What must a faithful same-scope survivor preserve and prove?
Endpoint line	What theorem or classifier verdict closes the component?

B Prediction Register Template

Register field	Required content
Prediction ID	Stable identifier, e.g. P7 Coupling/RG transport.
Target sector	Gauge, matter, Higgs, coupling/RG, flavor/generation, BSM/EFT, or universal.
Claim type	Future or submitted claim type governed.
Expected disposition	Carrier, representative, measurement, transport, route, mixing, projection, extension, update, burden, or failure.
Non-promotion rule	What skin cannot self-promote into primitive standing?
Falsifier	What exact same-scope primitive survivor would defeat the prediction?
Evidence route	Which experiment, model comparison, formal certificate, or sector theorem could test it?
Current status	Open, candidate, sector-proven, formally audited, or closed in SM-6.

C New-Physics / BSM Audit Template

Audit field	Question
Claim description	What new physics is being claimed?
Empirical surface	Is the claim a measurement anomaly, resonance, EFT coefficient, decay route, threshold, or detector pattern?
Model surface	Is the claim a new field, symmetry, particle, sector, compositeness proposal, or unification structure?
Same-scope test	Does it preserve D_{SM} and the carrier role bundle?
Descent test	Can it be classified as representative, measurement, transport, route, mixing, projection, extension, or skin novelty?
Target-update test	Does it change carrier inventory, invariant bundle, source state, admissibility boundary, or regime?
Prediction match	Which prediction-register item applies?
Falsification status	Does it rise above burden and close as a same-scope primitive survivor?
Verdict	Derivative status, target update, counterexample burden, failed primitive, or carrier revision.

D Candidate Theorem Inventory

Item	SM-1 status	Missing premise	SM-6 upgrade if closed
Skin Identification Lemma	closed criterion/lemma	local sector application	used in sector corollaries
No Primitive Skin Promotion	closed theorem form	skin eligibility for local object	global non-promotion theorem over certified sectors
Non-Ad-Hoc Classifiability	closed totality-or-failure theorem	P0 survival under future pressure	future-claim classifiability theorem
Fixed-Grammar Totality	conditional corollary	grammar adequacy for submitted claim	remains fixed-grammar theorem
SM Carrier Normal Form	closed under fixed grammar	grammar adequacy / local status data	global normal-form corollary
No Same-Scope Primitive Escape	sector-relative schema	sector carrier closures	global theorem
BSM Audit Closure	conditional schema	BSM/EFT audit closure	SM-6 corollary
Generation Cardinality Burden	high-risk sector target	role-cardinality closure	theorem or explicit open burden

Residual Primitive Deletion	SM-6 endpoint target	all sector closures	SM Carrier Exhaustion Closure Theorem
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E Worked Classification Certificates

Object	False primitive deleted	Expected status	Serious falsifier
Higgs	scalar, vev, potential, pole mass, hierarchy, vacuum branch, extra scalar name	$C_H = H_{\text{br/fix}}$, with skins and extension dispositions	Same-scope Higgs primitive outside C_H
Photon	field quantum, polarization, detector response	charge-response/gauge-order carrier with representative/route skins	new same-scope EM primitive outside charge response
Gluon	free asymptotic particle label, jet, color-flow diagram	non-asymptotic color-transport standing	free primitive gluon under fixed D_{SM}
Quark	isolated-particle skin, constituent/current mass label	color-bearing matter standing	free primitive quark under fixed D_{SM}
Coupling/RG	bare coupling, running crossing, scheme coordinate	transport/calibration ledger	coupling value generates primitive standing alone
CKM/PMNS	matrix entry, angle, phase, basis convention	flavor-transition coordinate skin / measurement route	matrix entry required as primitive transition identity
Fourth generation	field duplication as primitive generation	role-cardinality burden / target update / failure	same-scope fourth generation preserving carrier closure
Sterile neutrino	label as primitive neutrino identity	route, extension, burden, or target update	independent same-scope neutrino carrier outside network
Dark photon	field-name similarity as primitive EM identity	projection/interface enrichment, burden, or target update	primitive EM carrier outside charge-response standing
EFT anomaly	coefficient as primitive object	projection, transport, or route stress; burden	EFT coefficient alone supplies primitive standing

F Lean 4 audit surface crosswalk

The local crosswalk formerly shown in this appendix is superseded by the public Lean 4 audit reference in Section 31. The manuscript-facing crosswalk is retained here as an audit map, not as a separate

axiom-bearing Lean file.

Audit node	Lean-facing reading
Submitted claim carrier	typed record for submitted object, asserted scope, and target boundary
Status classifier	finite status algebra for carrier, representative, measurement, transport, route, mixing, projection, extension, target update, burden, and failure
No-promotion theorem	theorem over declared skin and certificate predicates; not a free-standing axiom
Carrier certificate	record of role closure, redescription invariance, licensing, descent exclusion, no-hidden-selector discipline, and local closure
Prediction register	typed indices for P0–P15 and their falsification conditions
Sector handoff	explicit support relation from SM-1 to SM-2–SM-6 and the generation-cardinality endpoint

G Referee Burden Table

Objection type	Has standing if...	Response if not...
Metaphysics objection	Shows classifier has no physical-domain input or no admissible target.	Reiterate status theorem over SM comparison arena.
Field ontology objection	Shows field notation independently carries primitive standing under fixed D_{SM} .	Grant representation; deny primitive field promotion.
New physics objection	Supplies target-preserving primitive survivor, not merely anomaly/model novelty.	Classify as measurement, route, projection, extension, burden, or update.
BSM objection	Preserves D_{SM} and closes new primitive carrier outside C_{SM} .	Mark BSM as audit-admissible, not protected.
Prediction objection	Shows a real claim cannot receive a registered status without ad hoc expansion.	State classification falsifiers and evidence tiers.
AASC dependency objection	Breaks a load-bearing theorem or invalidates specialization.	Treat dependency as declared proof class.
Carrier speculation objection	Shows SM-1 asserted final sector closure prematurely.	Mark carrier row as candidate until sector closure.

H Expanded Prediction Register

H.1 P0: Universal future-claim sorting

P0 is the master prediction. Its claim is that future Standard Model-associated novelty is not an untyped residue. A future claim may be difficult, empirically surprising, or theoretically disruptive, but it must still be submitted as something: a measurement, route, projection, field label, model surface, symmetry proposal, threshold, or carrier assertion. Once the surface is declared, the classifier has a route. P0 would fail only if a real Standard Model-associated claim preserved the target and nevertheless occupied no classifier status without adding a new status ad hoc.

P0 field	Content
Governed claims	all future SM-associated claims
Expected disposition	carrier, representative, measurement, transport, route, mixing, projection, extension, target update, burden, or failure
Non-promotion rule	chronological novelty does not self-promote
Falsifier	a real same-scope claim cannot be classified without ad hoc expansion
Evidence route	future anomaly/model/measurement audit; formal classifier stress test

H.2 P1: No same-scope BSM primitive escape

P1 is the BSM-specific version of primitive closure. It does not say BSM proposals are useless or false. It says that BSM status does not automatically carry protected primitive standing inside fixed D_{SM} . A proposal adding a field, symmetry, parameter, sector, or threshold must either descend to the carrier network, function as a registered derivative status, remain burdened, fail primitive promotion, or update the target.

P1 field	Content
Governed claims	extra scalars, vectors, fermions, hidden sectors, supersymmetry, GUTs, composite Higgs, sterile sectors
Expected disposition	descent, extension disposition, target update, burden, or failure
Non-promotion rule	model name, field label, symmetry name, or sector label does not self-promote
Falsifier	same-scope BSM primitive outside C_{SM} with closed certificate
Evidence route	BSM audit matrix plus sector closure comparison

H.3 P2: Skin-level new physics remains possible

P2 protects the project from empirical overclaim. New observations may appear as measurement witnesses, route novelties, projection novelties, transport corrections, EFT displays, anomalies, or extension dispositions. P2 predicts that such novelty will not by itself produce a new primitive. It may escalate burden or trigger target-update tests.

H.4 P3: Target-updating new physics is separated

P3 states that discoveries changing carrier inventory, regime, source state, invariant bundle, or admissibility boundary are not same-scope survivors. They update the target. This is not a way of dismissing

them. It is a way of preserving the logical difference between discovering a new domain and finding an unlicensed primitive inside the old one.

H.5 P4: Higgs-sector deviations

P4 is inherited from the Higgs arc. Higgs-sector deviations may be measurement, route, projection, extension, target update, burden, or failure. They do not automatically create a new primitive Higgs ground. The primitive Higgs status is $C_H = H_{\text{br/fix}}$, while scalar notation, vev language, potential, pole mass, vacuum-route language, and extension language are licensed downstream statuses.

H.6 P5: Quark confinement standing

P5 predicts that quarks do not acquire primitive isolated-particle standing under fixed SM scope. Quark standing is color-bearing matter standing under confinement-compatible discipline. A free quark would matter only if it carried same-scope primitive Standard Model standing, not merely if a route or projection suggested unusual color behavior.

H.7 P6: Gluon non-asymptotic standing

P6 predicts that gluons classify as color-transport carrier standing rather than free asymptotic primitive particles. Perturbative labels, jets, and color-flow diagrams are derivative. A free primitive gluon under fixed D_{SM} would be a severe falsifier.

H.8 P7: Coupling/RG transport

P7 predicts that coupling values, bare parameters, running crossings, matching conditions, and unification plots are transport/calibration ledgers unless a new carrier certificate is supplied. This prediction is important because many high-energy model comparisons rely on visually or algebraically striking RG behavior. The classifier treats such behavior as evidence, route, or transport pressure, not primitive standing by itself.

H.9 P8: CKM/PMNS matrix non-primitivity

P8 predicts that matrix entries, phase conventions, fitted angles, basis choices, and oscillation-route summaries are not primitive transition identity. They are transition-coordinate skins, measurement witnesses, or transport statuses unless a carrier-indexed transition certificate closes.

H.10 P9: Generation cardinality burden

P9 is deliberately marked high-risk. The generation question cannot be solved by saying that a fourth generation is “just another copy.” A same-scope generation addition must satisfy role-cardinality closure while preserving the representation-family ledger, mixing-coordinate dimension, Yukawa-realizability, anomaly/vacuum structure, and boundary bookkeeping. If it changes the target, it is target update. If it is merely duplicated field syntax, it fails primitive promotion.

H.11 P10: New scalar/vector/fermion claims

P10 covers the common particle-discovery surface. A new field label does not automatically supply same-scope primitive standing. It may be a representative, enrichment, route, projection, target update, burden, or failure. The falsifier is intentionally strict: field-label novelty alone must be shown to supply primitive standing under fixed D_{SM} , outside C_{SM} , with derivative statuses excluded.

I Expanded Carrier Register

I.1 Gauge-order carrier

The gauge-order carrier is not the gauge potential. It is the standing-preserving local order and admissible interaction-structure role that licenses gauge potential notation, generator charts, propagators, ghost displays, gauge-fixing choices, and local redescription. Its sector proof must show that gauge-related presentations are skins of one carrier role rather than multiple primitive objects.

I.2 Charge-response carrier

The charge-response carrier licenses charge numbers, currents, detector signatures, Coulomb-like displays, and representation-indexed response. The primitive object is not the numerical charge label. The number is a selector or display within a charge-response structure. SM-4 must separate charge-lattice rigidity from coupling-value derivation.

I.3 Color-transport carrier

The color-transport carrier is the standing role behind non-Abelian color response under confinement-compatible discipline. It licenses color labels, perturbative gluon notation, jets, flux tubes, color-flow diagrams, and parton-route language. Its central deletion is the primitive isolated quark/gluon skin under fixed D_{SM} .

I.4 Weak-chiral carrier

The weak-chiral carrier is the standing role behind weak chiral transition and charged/neutral response. It licenses W/Z notation, weak current labels, pole masses, widths, weak mixing displays, and electroweak route language. SM-2 must prevent W/Z mass or width values from being treated as primitive weak identity.

I.5 Matter-representation carrier

The matter-representation carrier is representation-indexed matter-response standing. It licenses spinor fields, generation labels, mass labels, flavor names, tracks, and route displays. SM-3 must show why matter identity is not grounded by spinor notation, mass value, track display, or isolated-particle skin alone.

I.6 Higgs bridge/fixation carrier

The Higgs carrier is the best-developed prototype. $C_H = H_{\text{br/fix}}$ mediates admitted order into deformation and mass-response standing. It licenses scalar notation, vev language, potential displays, pole mass, vacuum route, hierarchy orientation, and extension language. The Higgs arc already shows how a sector can preserve ordinary physics language while deleting false primitive promotions.

I.7 Flavor-transition carrier

The flavor-transition carrier is the standing role behind admissible transition across bases, phases, and measured routes. It licenses CKM/PMNS matrices, mixing angles, phases, oscillation baselines, and CP displays. The central danger is treating a matrix entry or phase convention as primitive transition identity.

I.8 Projection-interface carrier

The projection-interface carrier is the downstream interface through which metric displays, EFT coefficients, detector topologies, low-energy models, Planck-ratio comparisons, and phenomenological displays receive status. It is especially important for BSM/EFT audits, because much new-physics evidence first appears as projection pressure rather than carrier change.

J Detailed SM-2 Through SM-6 Handoff Receipts

This appendix now records the sector handoff as a receipt ledger rather than as a prospective writing plan. Each downstream paper receives the SM-1 classifier, fixes its local carrier/status package, and returns a scoped closure theorem or a declared burden/target-update branch.

Sector	Received from SM-1	Returned closure role
SM-2 gauge carriers	classifier, same-scope test, gauge-order payload, gauge skin atlas, and prediction register	photon, gluon, and weak-boson standing through gauge-order, charge-response, color-transport, and weak-chiral carrier roles, with derivative gauge displays typed as registered statuses
SM-3 matter carriers	matter-representation role, no-promotion discipline for spinor notation, mass labels, tracks, flavor labels, and route displays	quark, charged-lepton, and neutrino standing through matter carriers under confinement-compatible, charge-response, neutral-weak, mass-response, and route-interface discipline
SM-4 CCR transport	charge-response language, coupling/RG skin discipline, and parameter non-promotion obligations	charge, coupling, scheme, matching, threshold, EFT, and RG displays typed as carrier, transport, calibration, projection, burden, failure, or target-update statuses

SM-5 flavor/mixing/generation	flavor-transition language, matrix-coordinate non-promotion, and generation-governance burden typing	CKM/PMNS, rephasing, CP-phase, eigenstate, oscillation, FCNC, BSM flavor, and fixed-envelope generation branches closed as flavor statuses or handed to the generation-cardinality endpoint
SM-6 global closure	master classifier, sector closures, Higgs endpoint, projection/interface discipline, prediction register, BSM audit matrix, and target-update separation	final residual deletion $\text{Survprim}_{D_{\text{SM}}}(\text{Attr}_{\text{SM}} \setminus C_{\text{SM}}) = 0$ under the assembled carrier package

K Publication Package Checklist

Package item	Required content	Status in this build
SM-1 manuscript	master classifier, normal form, theorem family, prediction layer, worked examples	included
Prediction register	P0–P10 split into classifier-risk commitments and sector predictions, each with falsifier/evidence route	recorded in the manuscript ledger
Carrier register	candidate carrier classes, derivative skins, proof status	recorded in the manuscript ledger
BSM/EFT audit matrix	claim type, expected status, falsification pressure	recorded in the manuscript ledger
Theorem inventory	theorem names, closure statuses, and upgrade paths	recorded in the theorem ledger
Lean audit reference	public v0.1.0 Lean 4 audit surface, DOI 10.5281/zenodo.21219417, plus non-canonical local crosswalk maps	included
Build script	reproducible PDF build command	included
Audit report	PDF info and render/preflight notes	included in project archive

L External Orientation: Standard Usage Being Reclassified

The internal AASC references carry the proof burden. External references are used here only to fix the conventional physics-facing vocabulary that SM-1 reclassifies. Standard quantum field theory and Standard Model presentations fix the usual field, gauge, representation, symmetry-breaking, and renormalization language [62, 63, 64, 65]. Effective-field-theory and BSM reviews fix the ordinary use

of EFT coefficients, thresholds, and beyond-Standard-Model model surfaces [66, 67, 68]. Standard flavor and CP-violation treatments fix the conventional role of CKM/PMNS matrix coordinates and phases [69, 70]. Gauge-ontology and theoretical-equivalence discussions help locate the representational, non-ontic use of gauge redundancy [71, 72].

These references do not replace the AASC proof. They prevent a simpler misunderstanding: SM-1 is not denying standard field-theoretic practice. It is assigning primitive, derivative, transport, route, projection, burden, failure, or target-update status to the familiar objects of that practice.

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